Formalizing Microentrepreneurs through Targeted Tax Systems: Evidence from Brazil

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Abstract

We study the effects of reducing formalization costs by exploiting the introduction of a simplified tax regime for microentrepreneurs in Brazil. Using novel administrative data and industry-level eligibility criteria, we show that a large reduction in entry and compliance costs significantly increased formal firm creation. Despite the increase in formalization, newly formal firms exhibit no growth or earnings gains. We then evaluate the new tax system's fiscal consequences, showing that the expansion in social security coverage generated costs that exceed the additional tax revenue collected by 2.56 times. These costs arise as microentrepreneurs gain access to subsidized contributory benefits after registering through the system, highlighting the broader social value of bringing microentrepreneurs into the formal sector even in the absence of positive government net revenue or improvements in firm performance.

Keywords: Informality, Microentrepreneurs, Taxes, Registration Costs, Social Security Coverage.

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1 Introduction

Most businesses in developing countries are small and informal (Jayachandran, 2020). High levels of informality raise concerns among policymakers due to reduced tax revenues, limited access to social security, and the strong association between informality and lower productivity and earnings (Levy, 2008). As a result, there is significant interest in policies that encourage formalization. One widely discussed approach is to reduce the costs of formalization. While appealing in theory, the empirical evidence on the effectiveness of this strategy remains mixed (Ulyssea, 2020), and debates persist over whether the key barrier is the fixed cost of entry or the ongoing costs of remaining formal. Despite this attention, less is known about the fiscal consequences of such policies—especially the trade-off between lower tax revenues and increased public spending through expanded social security coverage of formalized individuals.

In this paper, we study the effects of one of the largest formalization programs in the world, leveraging novel administrative records and household survey data that capture both formal and informal labor markets. The program, called *Microempreendedor Individual* (MEI), was launched in Brazil in July 2009 and introduced a special tax regime that substantially reduced formalization costs for eligible microentrepreneurs. It simplified bookkeeping requirements, lowered tax rates, and eliminated registration fees. Prior to the program, the costs of registering a formal firm were estimated at 2,000 BRL—roughly four times the minimum wage—and required about 120 days. After full implementation, eligible individuals could register under MEI in a single day and at virtually no cost.

Our analysis focuses on three core dimensions. First, we examine how the reduction in formalization costs affected firm registration and the number of active formal firms among microentrepreneurs, and use the phased rollout of the program to shed light on the relative importance of different mechanisms—namely, entry costs versus tax rates. Second, we document a substantial increase in formalization among microentrepreneurs, but find no corresponding improvements in earnings or firm growth. Third, we evaluate the fiscal implications of the policy by quantifying both the increase in tax revenue and the additional costs associated with expanded social security coverage.

We begin our analysis by examining the effects of the MEI program on formal firm entry and activity. To identify causal effects, we exploit the program's industry-level eligibility criteria, defined at the 7-digit industry classification (CNAE). Our empirical strategy compares outcomes for eligible and ineligible industries before and after the program's implementation using a difference-in-differences design. We construct a monthly panel of industries using comprehensive firm-level administrative data from the national business registry (CNPJ), which allows us to track firm creation, and closure for all formal businesses over time.[1]

[1] Our data differs from previous work that relied on employer-employee matched data and only observed microentrepreneurs with formal employees. Rocha et al. (2018) uses RAIS, a matched employer-employee dataset organized by the Ministry of Labor, to estimate the effects of the MEI program but underestimates its impact on formal firm registration and activity, as MEI firms without employees are not required to report to RAIS.

We find that the MEI program led to a large increase in firm registration and the number of active formal firms. Our estimates indicate that the program caused the number of active formal firms to increase by 86% in eligible industries by December 2015. These effects are driven entirely by microentrepreneurs entering the MEI tax system, with no change in the number of formal firms outside the MEI regime. The timing and sharpness of these effects are consistent with a causal interpretation. Our granular administrative data allows us to document the parallel trends prior to the program and to precisely attribute the observed growth to firms entering under the MEI regime.

The phased structure of the MEI program allows us to discuss the mechanisms driving these effects. In its initial phase, beginning in July 2009, the program simplified taxes and bookkeeping requirements but retained substantial registration costs. In February 2010, a second phase introduced a fully integrated online registration system that eliminated in-person bureaucracy and virtually all entry costs. Finally, in May 2011, the government implemented a significant tax cut, reducing the flat monthly payment from 11% to 5% of the minimum wage. We compare responses across these phases to distinguish the effects of lowering entry barriers. Our findings indicate a sharp increase in firm registration after the drop in registration costs, suggesting that the burden of entry costs was an important barrier to formalization.

We then move beyond firm registration to better understand the program's impacts on formalization and microentrepreneurs' earnings, using household survey data that captures both formal and informal labor market activity. We draw on the nationally representative PNAD survey, which provides detailed information on individuals' employment status, social security contributions, and earnings. To implement our empirical strategy, we map PNAD's industry classifications to the program's eligibility rules and construct an industry-by-year panel. This allows us to compare trends in formalization and income across eligible and ineligible sectors over time using a difference-in-differences design.

We show that the program substantially increased formalization among microentrepreneurs but had no discernible effects on their earnings. Our difference-in-differences estimates indicate that, by 2015, the number of formal microentrepreneurs in eligible industries had grown by nearly 80% relative to ineligible industries. In contrast, average labor income among microentrepreneurs remained flat, with no significant differences between eligible and ineligible groups over time. At the aggregate level, the share of formal microentrepreneurs rose by 15 percentage points over six years—an increase of 79% from the baseline—while the total number of microentrepreneurs remained stable, indicating that the program induced formalization rather than shifts in occupational choice.

We complement these results with descriptive evidence from our administrative data, which allows us to track the full universe of MEI firms over time, including detailed information on ownership, employment, and transitions into and out of formal labor market ties. We find no evidence that the observed increase in formalization is driven by individuals gaming the system—such as formally employed workers reclassifying themselves as MEI to reduce tax liabilities.

Using linked employer-employee records, we show that the vast majority of MEI registrants were not previously employed in the formal sector, and transitions out of formal jobs at the time of MEI registration are limited and gradual. Furthermore, we document that newly formalized microentrepreneurs exhibit minimal firm growth. Four years after registration, 97% of MEI entrepreneurs remain non-employers, and even among those that hire at least one worker, the 90th percentile employs no more than five workers.

The final part of our analysis assesses the fiscal implications of the MEI program. While policies that encourage formalization are often motivated by the prospect of increased tax collection, they may also expand access to contributory benefits, generating additional fiscal costs. Using administrative records on MEI contributions and benefit receipts from the Federal Social Security Agency (INSS), we quantify both the revenue raised through the program and the costs associated with new social security claims. Although the number of contributors grew to over 3.3 million by 2015, total annual revenue remained under 800 million BRL. This modest result reflects not only the program's deliberately low tax rate, but also limited compliance, with only about 55% of expected monthly contributions actually paid.

Despite the modest revenue gains, the program significantly expanded access to social security benefits among microentrepreneurs. We estimate that the number of benefit recipients in this group increased by 56% relative to other occupations over the same period. This increase is driven primarily by maternity benefits and health-related pensions—benefits that the existing literature has shown to have important positive effects on recipients and household welfare.

Comparing the increase in revenue from tax collection with the increases in costs driven by higher social security coverage, we find that the program generated costs approximately 2.56 times greater than the revenue it brought in, resulting in a net fiscal loss of over 2 billion BRL in 2015, around 0.02% of the total government revenue in the year.

Our results indicate that the targeted tax system—by substantially reducing the costs of formalization—was effective in moving microentrepreneurs from the informal to the formal sector. On the one hand, the reform did not generate increases in fiscal capacity or meaningful productivity gains among formalized entrepreneurs, common goals of formalization policies. On the other hand, it successfully incorporated a substantial number of workers into the social security system, suggesting important welfare benefits for a vulnerable population.

To evaluate this trade-off in a unified and policy-relevant way, we assess the program through the lens of the Marginal Value of Public Funds (MVPF) (Hendren and Sprung-Keyser, 2020), which compares beneficiaries' willingness to pay for the policy to its net cost to the government. On the benefit side, we measure willingness to pay as the value of the social security benefits received by newly formalized microentrepreneurs, assuming a one-to-one valuation of transfers. On the cost side, we compute the net fiscal cost of the program as the increase in social security expenditures induced by the MEI program minus the additional tax revenues generated through contributions by new entrants into the formal system. Although the program does not generate large gross revenues, the fact that microentrepreneurs would have remained informal in the absence of the reform implies that these revenues would not have been collected otherwise. Combining these elements, we estimate an MVPF of 1.64, indicating that the willingness to pay of beneficiaries exceeds the net cost to the government.

This paper relates directly to the large literature that studies the causes of informality (La Porta and Shleifer, 2014). A central discussion in this literature has been about how entry costs and taxes affect informality. In recent literature reviews, Ulyssea (2020); Ulyssea et al. (2025) argue that economy-wide programs aimed at simplifying registration and reducing monetary entry costs tend to have limited effects on firm registration.

Our findings provide a counterpoint to this view. Combined with existing evidence, we provide a more nuanced, but very intuitive, positive relationship between (i) the effects of programs that implement and reduce the monetary costs of registering a firm and (ii) the level of formalization cost reduction. Our results are consistent with evidence from Peru, Benin, and Malawi (Mullainathan and Schnabl, 2010; Benhassine et al., 2018; Campos et al., 2023). Although these different contexts had their own idiosyncrasies, overall, the studies show that substantial policy reductions or experimental interventions in formalization costs had large effects on firm registration. Meanwhile, studies that find smaller effects—such as in Sri Lanka (De Mel et al., 2013) and Mexico (Bruhn, 2011, 2013)—focused on more modest reductions in administrative burdens.[2]

The major exception to this pattern is the study of the MEI program by Rocha et al. (2018), which plays a central role in the argument of Ulyssea (2020). Their findings suggest limited effects of the MEI program, and they argue that the observed impacts are driven primarily by tax reductions, rather than by substantial simplification or large reductions in entry costs. We show that using more comprehensive administrative data allows us to capture the full response to the targeted tax system, particularly among non-employer firm owners, which proves critical for correctly assessing the aggregate effects of the MEI program.[3] By revisiting the Brazilian evidence, we not only improve the discussion surrounding the MEI program, but also help reconcile findings across countries by emphasizing the importance of the magnitude of cost reductions. This allows us to link our findings to earlier work that posits this relationship, such as Djankov et al. (2002), and is also consistent with structural estimates in the literature (Ulyssea, 2010, 2018).[4]

[2]The SIMPLES tax system in Brazil has also promoted a debate on the effects of taxes on informality and development (Monteiro and Assunção, 2012; Fajnzylber et al., 2011), but recent work shows that conclusions are often taken from insufficient data (Piza, 2018).

[3]Appendix A provides a detailed discussion of the administrative data sources, their coverage, and why they are particularly well suited to capture these margins of adjustment. Appendix B provides a detailed comparison with Rocha et al. (2018), discussing their identifying variation and showing how both the magnitude of their estimates and their mechanism analysis are sensitive to specification choices.

[4]Two subsequent papers also study the MEI program. Pereda et al. (2022) focuses on gender differentials using household survey data and finds that women respond more than men to incentives, a pattern we also document. Carvalho (2024) examines the implications of the MEI program through a structural model of size-based taxation, adopting the same administrative data infrastructure introduced in our paper. Their analysis focuses on different margins and does not examine underlying mechanisms, social security benefits, or firm growth and employment

We also contribute to a broader debate about the value of formalization as a development strategy. A long tradition views informality as a key barrier to growth and allocative efficiency (De Soto, 1989; Busso et al., 2012; Dix-Carneiro et al., 2021; Zárate, 2019). However, a growing empirical literature finds limited economic gains from formalization at the micro level (De Mel et al., 2013; Benhassine et al., 2018). Our results are consistent with the latter: we find that formalized microentrepreneurs rarely expand, hire workers, or increase earnings.

Lastly, we provide unique evidence on an often understudied dimension of formalization policies, their effect on access to social security coverage. While much of the literature focuses on firm behavior and tax compliance, much less is known about how these policies expand entitlement to contributory benefits. This gap is particularly relevant in light of Levy (2008); Levy and Schady (2013), who emphasizes how segmented welfare systems in Latin America reinforce labor market dualism. We provide direct evidence on this channel by showing that the MEI program substantially increased access to pensions and other benefits among microentrepreneurs. However, the associated fiscal cost—driven by low contribution rates and a subsidized tax structure—was more than twice the revenue collected. These findings underscore the need to evaluate formalization policies not only through their effects on firm behavior, but also through their fiscal and redistributive implications.

2 Institutional Setting

As in many developing countries, microentrepreneurs constitute a large share of the Brazilian workforce. Defined as self-employed individuals or employers with at most one employee, they represented 22% of all workers aged 18 to 60 in 2008, according to the nationally representative household survey.[5] Yet only 17% of them contributed to the tax authority, compared to a 65% formalization rate among employees. Moreover, during the first five years of the *Partido dos Trabalhadores* (Workers' Party) federal administration, there was no measurable decline in microentrepreneur informality. In contrast, formalization among employees increased by seven percentage points over the same period.[6]

In December 2008, the Brazilian federal government launched the *Microempreendedor Individual* (MEI) program, which began operating in selected states in July 2009. The program aimed to increase the formalization rate of microentrepreneurs and, more broadly, to promote economic gains among newly formal firms. It introduced a new legal category within the tax system, allowing eligible microentrepreneurs to register as MEI firms. The program was created by Complementary Law No. 128/2008, which amended the 2006 National Statute for Micro and Small Enterprises (Complementary Law No. 123/2006) by adding Articles 18–A to 18–C.

transitions using administrative data, which play a central role in our evaluation.

[5] *Pesquisa Nacional por Amostra Domiciliar* (PNAD), conducted annually by the Brazilian Institute of Geography and Statistics. See below for details.

[6] The Workers' Party frequently advocated for labor formalization. Reducing informality was an explicit government objective in official platforms during the 2002 and 2006 presidential campaigns.

These articles defined the eligibility criteria, tax obligations, and benefits associated with MEI registration.

In this section, we provide a detailed explanation of the benefits associated with formalization through the MEI tax system, the eligibility criteria for the program, the differences in costs relative to the previous option, and provide a summary of the different phases of the program. We end the section with a description of our main data sources.

2.1 Benefits of Registering

The benefits that microentrepreneurs receive when registering their businesses through the MEI tax system are the same as in the other options of firm registration.

Under the statutory framework established by the National Statute for Micro and Small Enterprises, MEI firms are legally classified as microenterprises. This legal status allows them to operate formally, issue invoices, participate in public procurement processes, and access financial services such as business banking and payment systems. Although they are not limited liability entities, MEI registration enables access to formal credit in the firm's name. Registration also brings the firm into compliance with basic legal and tax obligations, reducing the likelihood of fines or penalties typically faced by informal operators and providing protection in the event of labor, tax, or sanitary inspections. In these contexts, the MEI is treated as a legally constituted business just as any other formal firm.

In addition to the firm-level benefits of formalization, individuals who register MEI firms become eligible for social security coverage through Brazil's public system. Upon registration, they are automatically enrolled in the *Regime Geral de Previdência Social* (RGPS) and contribute to the system via the *Instituto Nacional do Seguro Social* (INSS), the federal agency responsible for administering social security benefits. MEI contributors gain access to retirement pensions, disability insurance, and maternity leave, provided they meet the eligibility requirements and have contributed for at least ten months.

Taken together, these features underscore the dual nature of formalization benefits. On the one hand, registration grants access to a set of firm-level potential advantages traditionally emphasized in the informality literature—such as greater access to markets, finance, and contracting opportunities—that can facilitate business expansion and improve entrepreneurial performance. On the other hand, MEI registration also generates individual-level gains by integrating microentrepreneurs into the social security system, thereby extending a bundle of social protection benefits that are otherwise inaccessible to informal microentrepreneurs.

2.2 Eligibility Criteria

Although MEI firms offer similar legal protections and access to social security benefits as other formal firms, they differ substantially from the existing system in terms of eligibility requirements and cost structure.

To register as a MEI firm, microentrepreneurs must meet four criteria: (i) The owner cannot be a partner or shareholder in any other firm, and each MEI firm must have only one owner; (ii) Annual revenue must be below a threshold, originally set at 36,000 BRL; (iii) The firm may employ at most one worker; (iv) The firm must operate in one of the industries deemed eligible by regulation.[7] These industries are defined according to the *Classificação Nacional de Atividades Econômicas* (CNAE), a standardized 7-digit code used by the federal government to classify economic activities for tax and administrative purposes.[8]

The set of activities eligible for MEI registration is concentrated in lower value-added sectors. The list of eligible industries was inherited from those already designated under the National Statute of Micro and Small Enterprises as qualifying for the SIMPLES tax regime, Brazil's first major effort to reduce the tax burden on small firms. Since MEI operates under the same statutory umbrella, it adopted the same sectoral classification, which was originally established a decade before the implementation of the program, in 1998. As a result, the selection of eligible industries was not shaped by contemporaneous trends in firm dynamics or sectoral growth, but rather reflects longstanding criteria designed to target small-scale, low-complexity economic activities, where regulatory oversight was deemed less necessary, given the limited scale and lower operational risk of these activities.

Eligible industries include personal services such as hairdressers, manicurists, and house painters; small-scale commerce such as street vendors and shopkeepers; and basic trades including bricklayers, electricians, and repair workers. Ineligible activities include professions that require formal credentials or licensing—such as lawyers, accountants, and physicians—as well as sectors associated with higher revenue, capital intensity, or regulatory oversight, such as the education sector, manufacturing, industrial services, and food processing. We illustrate the variation in eligibility across broad sectors in Table A1, which reports the share of ineligible 7-digit CNAEs by the aggregate industry classification also derived from the federal government. To provide more detail on the specific types of activities included in each group, Table A2 lists the ten 7-digit CNAE codes with the most establishment registrations in 2008 among both eligible and ineligible industries.

2.3 Changes in Formalization Costs

The costs of formality were substantially lower for those registered under the MEI tax regime. These cost reductions occurred along three key dimensions: entry costs, taxes, and bookkeeping requirements.

Before the MEI program, the lowest-cost option available to a microentrepreneur seeking

[7] Eligibility is defined by Resolution No. 58 of the *Conselho de Gestão do Simples Nacional*, the federal body responsible for coordinating the rules of simplified tax systems.

[8] The first two digits represent a broad sector, the next three provide a narrower classification, and the final two define the most specific activity. For example, 4721-1/02 is the CNAE code for bakeries: 47 refers to all retail activities, 472 to retail of perishable goods, 4721-1 to retail of bread, milk, and similar products, and 4721-1/02 specifically to bakeries.

formalization was the SIMPLES tax system.[9] SIMPLES features tax brackets that vary by industry and firm revenue. In what follows, we compare the costs faced by MEI firms to those of SIMPLES firms operating in the services sector within the same revenue range. These differences—across taxes, entry procedures, and compliance requirements—are discussed below and summarized in Table 1.

Taxes: MEI firms pay a flat tax calculated as a percentage of the minimum wage. From the program's launch until April 2011, the tax rate was 11% of the minimum wage; beginning in May 2011, it was reduced to 5%.[10] In contrast, comparable firms in the SIMPLES tax system paid a rate of 4% on their gross revenue.

Because the two systems are based on different tax bases—minimum wage versus revenue—it is not straightforward to compare their effective tax burdens.[11] Nonetheless, we can approximate the relative burden of MEI taxation using household survey data. In 2009, the minimum wage was 465 BRL, implying a monthly MEI tax of 51.15 BRL. The median earnings of informal microentrepreneurs at the time were 600 BRL,[12] suggesting an implicit tax rate of 8.5% on profits. By 2011, when the tax rate was reduced, the minimum wage had risen to 545 BRL, lowering the flat monthly tax to 27.25 BRL. Median earnings of informal microentrepreneurs in 2011 were 760 BRL, implying an effective tax rate of approximately 3.5%. These figures indicate that the initial MEI tax rate was relatively high for a large share of microentrepreneurs, while the subsequent cut made MEI substantially more favorable—often lower than the equivalent SIMPLES rate.

Bookkeeping Costs: The flat tax structure of the MEI system substantially reduces bookkeeping requirements. MEI firms are not required to report the sources of their revenue or input costs and are exempt from submitting records reviewed by a certified accountant. In contrast, firms in the regular tax system must provide monthly accounting statements to the tax authority, a process that is often time-consuming and administratively burdensome. Regulations further require that these records be signed by an internal or external registered accountant. For microentrepreneurs, such services can be costly—sometimes exceeding the value of the taxes owed.

Registration Costs: Prior to the MEI program, registering a firm under the SIMPLES tax system required navigating a complex bureaucratic process involving federal, state, and municipal authorities. At the federal level, entrepreneurs had to register their firm with the national

[9]See Piza (2018) for a detailed overview of the SIMPLES regime.

[10]The rate reduction was introduced by *Medida Provisória* 529/2011 (later converted into law), which took effect on May 1st 2011. *Medidas Provisórias* are executive decrees issued by the President that take immediate effect without prior congressional approval or discussion, although Congress must subsequently vote to convert them into permanent law.

[11]This challenge is compounded by data limitations. In surveys that include both formal and informal microentrepreneurs, we typically observe earnings (likely equivalent to profits), but not revenue. This prevents direct estimation of the elasticity of tax rates on outcomes like formal firm registration or informality.

[12]Author's calculations using PNAD 2009.

tax authority. At the state level, registration with the *Junta Comercial* (Trade Board) and the state tax authority was required. Municipalities added further requirements, including business licenses and, depending on the activity, approvals from the fire department, health agency, and environmental regulators. On average, this process cost around 2,000 BRL and took approximately 120 days to complete.[13]

When the MEI program began in July 2009, the registration process initially resembled that of other systems.[14] Individuals had to complete approximately 40 online forms, print and sign documents, and submit them in person to the Trade Board. This would initiate the state-level process, which proceeded until the firm was formally approved. Licenses and permits were only waived if local governments passed enabling legislation.

In February 2010, the federal government dramatically simplified the process. A new online platform allowed individuals to register by providing only their name, social security number, date of birth, and intended industry of operation.[15] Registration fees were eliminated, in-person appointments were no longer required, and the need for licenses and permits was removed in most cases. Firms could now complete the process in a single day and become immediately compliant with tax and regulatory obligations.

Table 1: Comparing Cost for Firms in MEI and SIMPLES Systems

	SIMPLES	**MEI**
Tax Rate	4% of Revenue	Flat Tax Rate Jul/09 - Apr/11 - 11% of M.W. May/11 - onwards - 5% of M.W.
Entry Costs	Avg. 2000 R$ process Different steps with at least 4 Government Agencies Average 120 days	Starting Feb/10 - Virtually Costless Integrated system in one Website Registration complete in 1 day
Bookkeeping	Necessary Bookkeeping Accountant is mandatory by Law	No necessary Bookkeeping

This table plots the differences in cost of entry and compliance to firms in the formal sector under the MEI system and the previous best alternative to microentrepreneurs (SIMPLES). The SIMPLES tax system has different tax schemes according to industry and revenue progressivity. In this table compare the costs of a MEI firm relative to SIMPLES firm in the services sector in the same revenue range of MEI firms.

[13]See this report for a detailed breakdown of registration costs for non-MEI firms in 2009. Estimates are consistent with the World Bank's Doing Business survey (World Bank, 2010). Figure A1 summarizes the steps involved.

[14]Requirements varied by firm type. This presentation outlines the documents required to register a MEI firm in 2009.

[15]See this announcement from the federal government detailing the launch of the new system.

2.4 Program Timeline Summary

We described the main institutional features of the MEI regime and documented the key implementation changes introduced during its early years. To organize these developments and facilitate interpretation of the program's effects, we summarize the rollout using a three-phase timeline. Although simplified, this structure highlights the major shifts in coverage, administrative procedures, and tax rules that shaped how microentrepreneurs experienced the program over time.[16]

- **Phase 1** – July 2009 to February 2010: Implementation began in selected states,[17] with the MEI tax rate set at 11% of the minimum wage. Registration still required a substantial number of procedural steps, including multiple online forms and in-person submissions across different government agencies.
- **Phase 2** - February 2010 until May 2011. The program is implemented in all states. Tax rate at 11% of the minimum wage. Simplified registration process, fully online, potentially one day process.
- **Phase 3** - May 2011 - onwards. Tax rate reduced to 5% of the minimum wage. Previous features apply.

2.5 Data

We draw on two primary data sources for our analysis, supplemented by additional datasets described in Appendix A.

Cadastro Nacional de Pessoa Jurídica (CNPJ): The CNPJ is the national registry of all formal firms in Brazil, including those registered under the MEI program. It serves as the main building block of our dataset on formal microentrepreneurs. For each firm, we observe a unique firm identifier, official name, date of registration, date of closure (if applicable), industry classification (at the seven-digit CNAE level), legal nature of the entity, address, and an indicator for whether the firm is registered under the MEI regime. Additionally, for 93% of MEI firms, we can recover the owner's name and social security number (*Cadastro de Pessoa Física*, or CPF), which enables us to track unique microentrepreneurs.[18]

The use of CNPJ is a key distinction relative to previous findings about the MEI program. Rocha et al. (2018) use the matched employer-employee RAIS data in their analysis. However, MEI firms that do not employ were exempted from reporting to RAIS. In turn, all

[16] Figure A2 provides a visualization of the timeline and also summarizes the major changes of the program.

[17] São Paulo, Minas Gerais, Distrito Federal, and Rio de Janeiro in July 2009; Rio Grande do Sul, Santa Catarina, Paraná, and Ceará in September 2009.

[18] This is due to a design choice introduced in February 2010, when MEI firms' registered names (*Razão Social*) were required to include the owner's full name and CPF. Table A3 shows the share of firms for which we recover this information.

firms—including MEI—must register in the CNPJ. In Appendix A, we have a detailed analysis of the structure and accuracy of the CNPJ data, and an explanation of the RAIS vs CNPJ differences. We show that only around 10-15% of MEI firms in CNPJ eventually report to RAIS - highlighting the importance of correct records in the analysis of the MEI program.

Pesquisa Nacional por Amostra Domiciliar (PNAD): Because the CNPJ data only captures formal firms, we rely on PNAD—a nationally representative household survey conducted annually by the Brazilian Institute of Geography and Statistics (IBGE)—as our primary source for informal microentrepreneurs. PNAD contains individual-level information on demographics, employment, income, and social security contributions. We use it as a repeated cross-section, as individuals are not followed over time.

We prefer PNAD over the *Pesquisa Mensal de Emprego* (PME), which has been used in Rocha et al. (2018); Pereda et al. (2022); Carvalho (2024), for different reasons. First, PNAD provides a more detailed industry classification, with roughly 210 five-digit categories, and IBGE offers a direct crosswalk to the fiscal CNAE codes used to define MEI eligibility. In contrast, PME includes only 90 industry categories and lacks an official crosswalk. Second, PNAD is nationally representative, while PME covers only metropolitan areas. In Appendix B, we replicate our key informality measures using PME data and provide a detailed discussion of the results of Rocha et al. (2018) and their use of PME data.

3 Effects on Formal Firm Entry and Activity

In this section, we examine the effects of the MEI program on formal firm registration and the number of active firms. We begin by outlining our empirical strategy, which exploits variation in industry-level eligibility. We then present the main results, explore potential mechanisms driving the observed effects, and address alternative explanations.

3.1 Empirical Strategy

To estimate the causal effect of the MEI program on firm entry and the number of active firms, we use a difference-in-differences approach that compares eligible and ineligible industries before and after the program's implementation. Industries are defined at the 7-digit CNAE level.

We aggregate firm-level data to the industry level and construct a monthly panel tracking firm entries, exits, and the stock of active firms. Treated industries are those eligible to register under the MEI regime. For our baseline estimates, we use all remaining industries as controls, and we conduct robustness checks using alternative control groups matched on pre-program characteristics. Our primary sample includes 1,323 unique CNAE industries, of which 371 are treated and 952 serve as controls.[19] The panel covers the period from January 2005 to December

[19]We exclude CNAE 9492800, which corresponds to political organizations, as electoral committees are legally required to register in this category during election periods.

2015.

In practice, we estimate the following difference-in-differences specification using this industry-month panel to recover the dynamic treatment effects:

$$Y_{ct} = \theta_c + \gamma_{s(c)t} + \sum_{t=01/2005}^{12/2015} \beta_t \cdot D_{ct} + \epsilon_{ct} \tag{1}$$

where D_{ct} is a dummy that takes value one in month t if the 7-digit industry is eligible. We normalize the coefficient β_t for t equals June 2009 to zero. Thus, the vector β_t captures how much our outcomes of interest vary between period t and June 2009 in eligible and ineligible industries. We also add fixed effects of aggregate sector x time, represented in the equation by $\gamma_{s(c)t}$. Aggregate sectors are grouped by the first 2-digits of the industry definition. This allows us to control for aggregate trends that are time-varying and possibly correlated with our treatment.

To summarize the effects across different phases of the program, we also estimate a pooled specification that interacts treatment status with indicators for the three program periods:

$$Y_{ct} = \theta_c + \gamma_{s(c)t} + \beta_1 \cdot P1_t \cdot D_c + \beta_2 \cdot P2_t \cdot D_c + \beta_3 \cdot P3_t \cdot D_c + \epsilon_{ct} \tag{2}$$

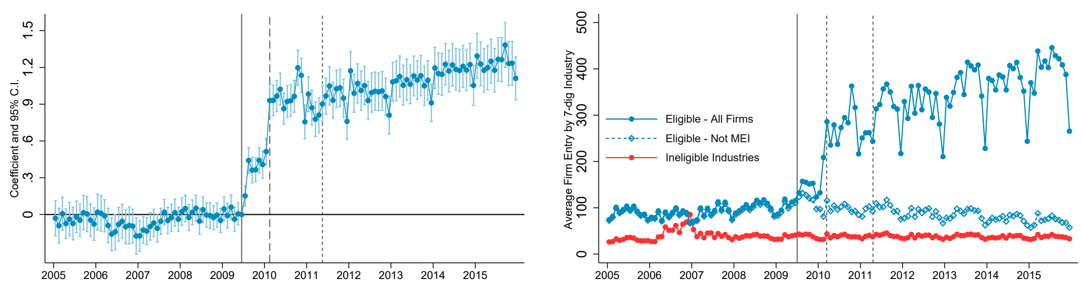
where $P1_t$ is a dummy that takes value one if the month is in the implementation phase between July 2009 and January 2010. $P2_t$ indicates that the month is between February 2010 and March 2011. And $P3_t$ indicates the period from May 2011 until December 2015. Thus, the coefficients $\beta_t \quad t \in \{1, 2, 3\}$ reflect how much the outcomes differ between eligible and ineligible industries in periods 1, 2, and 3 relative to the pre-MEI program period. As eligible and ineligible industries differ in their baseline characteristics, we interpret the estimated coefficients β_t as the average treatment effect on the treated (ATT).

Our identification strategy relies on the assumption that, in the absence of the MEI program, eligible and ineligible industries would have experienced similar trends in outcomes. Although not testable, the parallel trends assumption is plausible in our setting for both institutional and empirical reasons. As discussed in Section 2, the eligibility criteria for MEI were based on industry classifications that predate the program by more than 10 years and were not directly related to differential trends in formality or business creation that would be confounded at the time of the introduction of the MEI program.

3.2 Estimates on Formal Firms' Entry and Number of Formal Firms

We begin by presenting our estimates of the MEI program's effect on formal firm entry, measured as the logarithm of firm registrations. One advantage of focusing on firm registration is that it is a flow measure and thus likely to respond more immediately to institutional changes than stock measures such as the total number of active firms. Combined with monthly data, this allows us to trace the program's impact with high temporal precision and examine how effects

Figure 1: Effects on Firm Entry



(a) Differences in Differences Estimates

(b) Avg. Firm Entry in Levels

This figure shows the effects of the MEI program on formal firm entry. In Panel A, we compute the number of firms registering in each CNAE in each month, and then calculate the average across eligible and ineligible CNAE codes. The dashed blue line shows the number of not MEI firms in eligible industries. In Panel B, we plot the vector of coefficients β_t and 95% confidence intervals from equation 1. The outcome is the logarithm of the number of firm registrations, and standard errors are clustered at the 7-digit CNAE level. In both panels, the sample consists of a monthly panel at the 7-digit CNAE industry level from 2005 to 2015. Vertical lines mark the beginning of each program phase.

vary across implementation phases.

We find a large and sustained increase in firm entry in eligible industries. Figure 1b shows the average firm entry in levels in eligible and ineligible industries, and Figure 1a presents estimates of β_t from equation 1.

Three main results emerge from Figure 1a. First, there is no evidence of differential pre-trends between treated and control industries, supporting the validity of the identification assumption. Second, program effects began immediately in July 2009 and increased during Phase 2, when the program was fully operational. The modest effects in Phase 1 are expected, as not all states were yet authorized to register MEI firms and the integrated website had not been launched. Third, we find no substantial change in registration behavior between Phases 2 and 3. If anything, additional growth appears only in 2014–2015, rather than immediately after the 2011 tax cut. We discuss the implications of these patterns for understanding program mechanisms below.

Figure 1b provides additional understanding about the estimated effects. After the program's implementation, the increase in registrations is concentrated in MEI-eligible firms. Non-MEI registrations remain stable, suggesting the increase is not driven by broader economic shocks or reclassification behavior. Moreover, ineligible industries show no signs of trend breaks around the program start, which helps rule out spillovers or strategic industry switching. Appendix Figure A3 replicates these patterns using log-transformed outcomes, and shows similar patterns.

Table 2 presents estimates from the pooled specification in equation 2, summarizing the program's effects across phases. In our preferred specification—including fixed effects for 2-digit industries interacted with month—firm registrations increased by 166% in Phase 2 and 200% in

Phase 3. Columns (3) and (4) report analogous results using the number of firms (in levels) as the outcome, yielding similar estimates.

Table 2: Estimates of Program's Effect on Firm Entry

	(1)	(2)	(3)	(4)
	Log(Firm Entry)		Firms Entry	
Eligible x Phase 1	.45***	.42***	60***	37***
	(.03)	(.036)	(9.3)	(7)
Eligible x Phase 2	1.1***	.98***	186***	122***
	(.045)	(.06)	(29)	(20)
Eligible x Phase 3	1.2***	1.1***	267***	191***
	(.055)	(.077)	(44)	(32)
Dep. Var. Mean of Treated	3.04	3.04	93.98	94.38
Month x 2 digit Ind. F.E.	No	Yes	No	Yes
Observations	143433	142404	143433	142404

This table presents estimates of coefficients $\{\beta_1, \beta_2, \beta_3\}$ from equation 2. Columns (1) and (2) use the logarithm of firm registrations as the outcome. Columns (3) and (4) use the number of firms (in levels). Standard errors are clustered at the 7-digit CNAE industry level.

Estimates on Number of Active Formal Firms: Next, we show the effects of the program on formal firm activity. We define a firm as active at period t if its date of entry was before t and its date of closure (if closed) was after t. If the firm had not closed until December 2015, it is defined as active from the moment it was registered until the end of our analysis period.

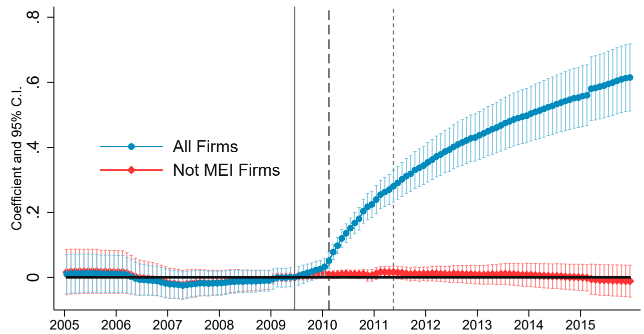
Our results are presented in Figure 2. It plots the estimates of model 1 for the log of the total number of active firms and the log of the number of active firms not in the MEI program. We find that, in eligible industries, the number of active firms increased by approximately 40% by December 2012 and by over 86% by December 2015[20]. These estimates are substantially larger than those reported in Rocha et al. (2018), who, using RAIS data, find only a 4% increase in the number of firms by 2012. As discussed, MEI firm owners without employees are not required to report to RAIS, which likely leads to a significant undercount in their analysis. The magnitude of the difference between the two estimates is aligned with the share of MEI firm owners without employees.

We also see that all the effect in the number of active firms is driven by MEI firms as the number of active firms not in the MEI program is stable throughout the whole period.[21] This is

[20] Computed as $\exp(0.62) - 1 = 0.86$

[21] We also run the same specification in equation 1 for the log of firms closing. We show the results in Figure A5. We see a large increase in firm exit in eligible industries after the start of the program, and this is driven fully by MEI firms. But coefficients are smaller than the ones for firm registration, which leads to the positive

Figure 2: Difference in Differences Coefficients on Active Firms



This figure plots the vector of coefficients β_t from equation 1 on the natural logarithm of the number of active firms in each 7-digit CNAE industry. The blue line shows the coefficients and 95% Confidence Interval of the regression using all firms as outcome whereas the green line shows the same statistics of a regression using only the number of Not MEI firms as outcomes. Vertical lines represent the start of the different phases of the program.

informative for two reasons. First, the absence of any change in not-MEI firm counts helps rule out the possibility that other industry-level shocks are driving the results. If such shocks were present, we would expect to see increases in both MEI and not-MEI firms. Second, the stability of non-MEI firms suggests that individuals are not simply shifting from other tax regimes, such as SIMPLES, into the MEI system. Since it is legal to close a SIMPLES firm and open a MEI firm, one might be concerned that such substitution could be driving the observed increase in firm registrations. However, if this practice were occurring at scale, we would expect to see a decline in the number of non-MEI firms.

Additional Checks: One potential concern is that the low cost of MEI registration may have induced microentrepreneurs to register without ever becoming active, thereby overstating the program's impact on formal firm activity. If many registrants quickly ceased operations or never began them, the observed increase in the number of active firms would exaggerate the true effect. To address this, we draw on data from the *Perfil do MEI* survey, a qualitative study conducted annually by SEBRAE, a federal agency supporting small businesses - we describe this data in detail in Appendix A. From 2012 to 2015, approximately 90% of respondents reported being currently active. In 2015, the survey further broke down the inactive group, revealing that one-third of them intended to start operations soon. These figures suggest that the CNPJ registry provides a reliable measure of formal activity and support the validity of our estimates. Summary statistics from the survey are reported in Table A6.

As an additional robustness check, we re-estimate our baseline difference-in-differences models using a matched sample of control industries. Specifically, we implement a one-to-one propensity score matching procedure without replacement, using 2008 industry-level characteristics: the

result on formal firm activity.

number of firm entries, the number of active establishments, and the share of firms with limited liability. This procedure yields a balanced sample of treated and control industries. Table A4 and Figure A9 show that the estimated effects remain very similar to those from our main specification using the full control sample.

Another potential concern is that our estimates might capture changes in enforcement rather than the effects of the MEI program itself—specifically, an increase in labor inspections, which the literature has identified as an important driver of firm formalization de Andrade et al. (2016); Ulyssea (2020); Ulyssea et al. (2025). If the rollout of MEI coincided with a surge in enforcement activity, our results could overstate the program's impact. However, Figure A8 shows that labor inspections were actually on a downward trend during the MEI implementation period. Using inspection data from Ponczek and Ulyssea (2022), we document both the total number of inspections nationwide and the average number of inspections among municipalities with at least one inspection in a given year. Beyond the aggregate decline, we are not aware of any reallocation of inspections toward MEI-eligible industries at the time of implementation that could plausibly bias our estimated treatment effects.

3.3 Discussion on Mechanisms: Taxes or Entry costs?

The results described above provide clear evidence that the MEI program had large effects on both the number of firms registering in the formal sector and the number of active formal firms. This finding challenges the prevailing view in the literature, summarized by Ulyssea (2020), that economy-wide programs aimed at simplifying registration and reducing monetary entry costs tend to have limited effects on firm registration.

In addition, our setting allows us to provide suggestive evidence on the mechanisms underlying the program's effects by comparing outcomes across its different phases. The discussion that follows should be interpreted as strongly suggestive rather than causal, as we do not employ a research design capable of isolating specific mechanisms. Even without causal identification, this analysis is important, particularly because earlier evidence from the MEI program has led to a prevailing view of the underlying mechanisms that our results substantially revise.

As discussed in Section 2, the program unfolded in three distinct phases. The first was an implementation phase, during which only a subset of states were eligible and entry costs remained substantial. In the second phase, the program operated nationwide but maintained a tax rate of 11% of the minimum wage. The third phase began when this tax rate was reduced from 11% to 5%. Rocha et al. (2018) argue that the program's effects emerge primarily after May 2011—corresponding to our Phase 3—and therefore conclude that neither the reduction in entry costs nor the administrative simplification had meaningful effects on formalization, attributing observed increases in formal firm registrations largely to the tax reduction. Our estimates provide clear evidence that the MEI program generated substantial formal firm entry during Phases 1 and 2, leading to a different interpretation of the timing and sources of the program's effects. Moreover, Appendix B shows that, even when using the same RAIS data as in Rocha et al. (2018), estimates that leverage the appropriate identifying variation also yield substantial effects in Phases 1 and 2.

While the program's timing across phases is informative, we remain cautious in attributing its effects to any single mechanism. Early impacts may reflect announcement effects or temporary increases in salience. The MEI program also introduced a fundamental shift in the tax base—from a revenue-based contribution under SIMPLES to a flat monthly payment. Moreover, reductions in bookkeeping requirements and administrative simplification may have further contributed to the observed patterns.

Even with this caution, the sharp increase in formal firm entry between Phases 1 and 2 provides important suggestive evidence. Two key changes occurred in February 2010. First, the program expanded to the entire country, eliminating geographic restrictions on eligibility. Second, the introduction of a fully integrated online registration portal substantially reduced the number of steps and interactions required to formalize.

To isolate the effects of the different entry months for different states, we provide separate estimates changing the level of aggregation to the industry $\times$ state $\times$ month level (rather than industry $\times$ month). We then divide the sample according to the month in which each state gained access to the MEI program, creating separate subsamples for early-adopting and later-adopting states. For each of these subsamples, we estimate our difference-in-differences specification independently. This setup allows us to compare how firm entry evolves around February 2010 within states that had already been exposed to the program for different lengths of time.

We find that formal firm entry rises sharply after February 2010 even in states where the MEI program had been available as early as July 2009. We present these results in Figure A6, which shows estimates of equation 1, and Table A5 which shows our pooled estimates from equation 2. For these early-adopting states, the only relevant institutional change occurring at that moment was the introduction of the simplified online registration system. This pattern provides compelling suggestive evidence that the reduction in entry and compliance costs was a key driver of the program's early effects.

4 Effects on Formalization and Microentrepreneurs' Earnings

Section 4 shows that the MEI program generated a large increase in formal firm entry, but by the nature of the data, those results speak only to outcomes observed within the formal sector. This naturally raises the question of who these newly formal microentrepreneurs are—whether they were previously informal, out of the labor force, or changing occupations—and how their labor market outcomes change once they formalize.

To address these points, this section draws on household survey data, which captures both formal and informal work, and uses additional administrative records that complement our previous findings. Together, these sources allow us to examine changes in overall formalization, earnings, and transitions from formal employment into formal microentrepreneurship.

4.1 Aggregate Patterns around MEI Program Implementation

We begin describing aggregate trends on the labor market and formalization during the period of our analysis of the MEI program. The patterns suggest that the increase in formal firm registration is driven by the formalization of informal microentrepreneurs, rather than by individuals entering the labor force or by workers changing their occupation to microentrepreneurship.

First, there are no significant changes in the share of microentrepreneurs in the working population, nor in the share of the adult population in the Labor Force. This is shown in Figures A10a, where we plot the share of adult population working and in Figure A10b, where we divide the working population in 4 main groups (Employees, Microentrepreneurs, Employers, and Others - namely unsalaried workers) and show how these groups evolve around the implementation of the MEI program. The share of microentrepreneurs among the working force remains constant around 21-22% between 2002 and 2015, and the labor force participation is constant at 70%.

In turn, the share of formal microentrepreneurs increased by 15 percentage points in the six years following the program's implementation.[22] This represents a 79% increase in the formality rate. Figure 3a shows that the share of formal microentrepreneurs was stable at around 18–19% from 2002 to 2008, before rising sharply after MEI was introduced.

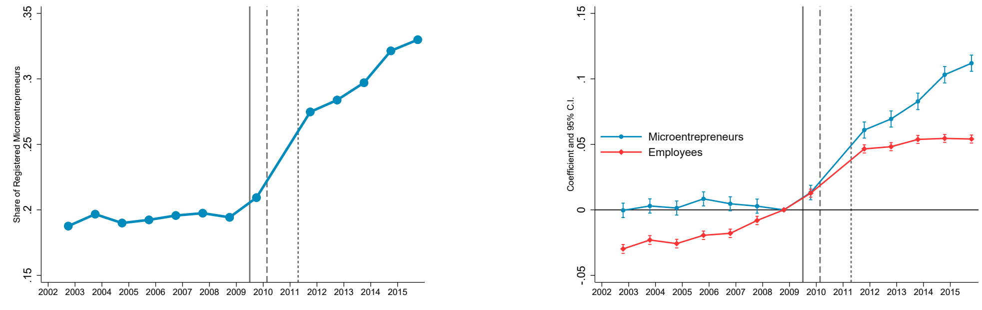
The composition of microentrepreneurs is not driving these changes. This is particularly relevant in the period between 2002 and 2015, when Brazil observed a demographic change, with a large increase in education levels. To investigate the effects of composition changes on formalization rates of microentrepreneurs, we residualized the probability of an individual being formal on a series of observable characteristics. We observe that the pattern controlling for observable characteristics in Figure 3b is very similar to the simple averages. There are no changes between 2002 and 2008, followed by a large increase in the share of formal microentrepreneurs after MEI is implemented.

As an additional piece of evidence, in the red series in Figure 3b, we replicate the same exercise but with a sample of employees. The 2000's decade was a period of formalization of employees (Haanwinckel and Soares, 2021). However, the increase in microentrepreneurs' formality does not appear to be concurrent with the trend of employee formality.

Heterogeneity: We decompose our sample in different ways to better understand how changes in informality rates differ across groups. We show these results in Figure A12, where each Panel shows a different sub-sample comparison. We find that women face a larger change in their informality rate than men. By 2015 female microentrepreneurs are more likely to be formal than male ones. One possible reason for why this is happening is because the MEI program grants

[22] In what follows, we exclude entrepreneurs in agriculture. The same patterns including agriculture are shown in Appendix Figures A11 and A13. In Appendix B, we also replicate aggregate trends using PME data, as in Rocha et al. (2018).

Figure 3: Share of Formal Microentrepreneurs and Employees



(a) Microentrepreneurs **(b)** Controlling for Observables

Microentrepreneurs are defined as individuals who declare themselves as self-employed or employers who hire at most one employee. We define a formal microentrepreneur as those who declare to be contributing to social security through their job. Panel A plots the share of microentrepeneurs between 18 and 60 years that are formal by year. Panel B plots the estimated year effects from a regression of formal status on time dummies and observable characteristics for all microentrepreneurs between 2002 and 2015. The specification controls for gender, race, log monthly income, a second-order polynomial in age, and fixed effects for schooling, industry, and state (with 2008 normalized to zero). Blue circles show results for a sample of microentrepreneurs and red diamonds show results for a sample of employees.

paid maternity leave to women who contributed to the tax authority for at least 10 months. When looking at race, white individuals have a slightly larger increase in absolute terms than nonwhite ones, although the percentage increase is higher for the nonwhite population, which starts from a significantly lower baseline. We also see that after the implementation of the MEI program, there is a large increase in the formality rate of microentrepreneurs with high school education and those in the middle of the earnings distribution.

4.2 Industry Level Analysis of Effects on Formalization and Earnings

A major feature of PNAD data, is that its granular industry definition allows us to conduct the industry-level analysis with appropriate adjustments, but exploring other outcomes in the data.[23] We aggregate the data into an industry x year panel. We calculate the total number of microentrepreneurs, their average earnings in the job, and the number of formal microentrepreneurs. We then use the crosswalk between fiscal CNAEs in which the MEI program eligibility was defined, and the *CNAE domiciliar*, the industry classification used in PNAD. With our panel organized, we estimate the following regression:

$$Y_{ct} = \theta_c + \gamma_t + \sum_{t=2002}^{2015} \beta_t \cdot D_{ct} + \Gamma X_{ct} + \epsilon_{ct} \tag{3}$$

[23] Given that PNAD has individuals' earnings, one could consider the possibility of a Regression Discontinuity Design around the cutoff of earnings eligibility. Unfortunately, as in many other household surveys, in PNAD, there is a clear reporting bias at round numbers for the earnings of self-employed and microentrepreneurs. This is generates an issue for a potential RDD as the eventual running variable is not smooth around the cutoff. In Figure A14 , we plot the earnings distributions of microentrepreneurs in 2008-2011 PNAD, where we can clearly see the bunching around the cutoff

where θ_c are *CNAE domiciliar* fixed effects, γ_t are year fixed effects, and D_{ct} is a indicator that takes value one for eligible CNAEs in period t. The main difference relative to equation 1 is that our panel is annual instead of monthly. We are also able to include time-varying controls such as race, gender, schooling, and the geographical composition of microentrepreneurs. The identification assumption in this model is the same as before: in the absence of the MEI program, eligible and ineligible industries would follow parallel trends. As before, eligible and ineligible industries are not the same, thus our coefficient is interpreted as an Average Treatment Effect on Treated industries.

Our main findings are summarized in Figure 4. In Panel 4a we plot the average number of formal microentrepreneurs in eligible and ineligible industries. In Panel 4b we show our estimates of the difference in differences model. To facilitate the interpretation, we divide our estimates by the baseline number of formal microentrepreneurs in treated industries in 2008.[24] Panels 4c and 4d show the same exercise, but using average wages as the outcome.

Our estimates indicate that MEI increased the number of formal microentrepreneurs in eligible industries by almost 80% by 2015. We observe that both eligible and ineligible industries follow similar trends prior to 2009, despite an initially higher number for eligible industries. However, after the program, we see a substantial increase in the number of formal microentrepreneurs in eligible industries, whereas we do not see the same pattern in ineligible ones. Importantly, in Figure A17 we replicate this exercise using the total number of microentrepreneurs (formal and informal) as the outcome. We observe no change, indicating that increases in microentrepreneurship in eligible industries are not driving our findings. This is consistent with the lack of changes in occupation composition described in aggregate patterns.

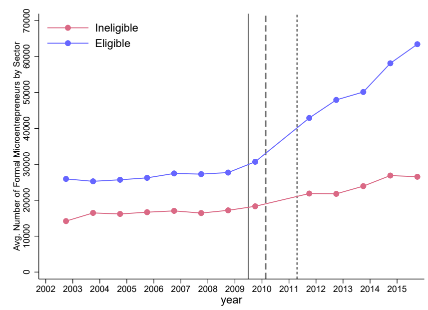
At the same time our results show that the MEI tax system did not have effects on microentrepreneurs' earnings in eligible industries. None of our estimated coefficients is statistically different from zero. Although the estimates are not very precise, the post-policy point estimates for β_t hover around zero. We also find no effects on the number of employees from firms in these sectors using PNAD data. This is presented in Figure A18.

These findings speak to one of the central motivations behind the creation of the MEI program - the expectation that formalization would improve the economic performance of microentrepreneurs. Several mechanisms could, in principle, have led to higher earnings once individuals registered as MEI. From a theoretical perspective, Ulyssea (2018) emphasize that informal firms often operate at a suboptimal scale because growing increases their visibility and the risk of enforcement; formalization relaxes this constraint and could allow firms to expand output, hire workers, and increase profits. In practice, the MEI program also enabled different ways in which microentrepreneurs could have grown. Registration permits the issuance of invoices, participation in public procurement, and integration into formal supply chains, all of which could expand demand for their services. Formal status may likewise improve access to credit markets, enabling investment and business expansion.

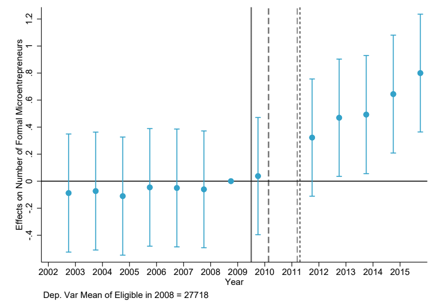
[24] We present our coefficients without dividing by the baseline in Figure A15.

However, the lack of effect on earnings and firm size are consistent with the literature. Recent literature reviews Ulyssea (2020); Ulyssea et al. (2025) argue that, despite some mixed results, most estimates of the effects of formalization find no statistically significant impact on various measures of firm performance, including sales, profits, and employment. Our findings add to this body of evidence, indicating that the potential channels that would increase firm performance due to formalization have limited effects.

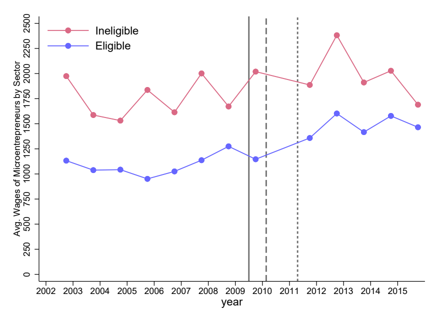
Figure 4: Effects on Formalization and Earnings of Microentrepreneurs



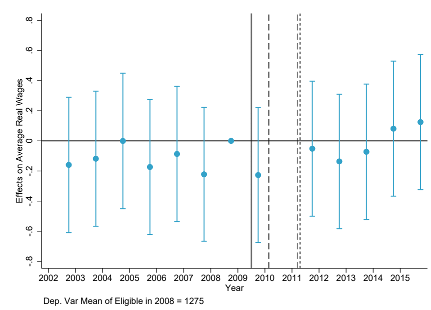
(a) Formal Microentrepreneurs - Trends



(b) Formal Microentrepreneurs - Estimates



(c) Average Labor Earnings - Trends



(d) Average Labor Earnings- Estimates

This figure shows formalization of microentrepreneurs across eligible and ineligible industries using PNAD data. Panel (a) shows the number of formal microentrepreneurs among eligible and ineligible industries. Panel (b) plots the coefficients β_t from equation 3 divided by the average of the outcome variable in 2008. The sample comprises all microentrepreneurs in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state. 2010 estimates are missing because PNAD was not collected in that year.

4.3 Following Microentrepreneurs Through Time

A distinctive feature of our data is that they allow us to track virtually all formal microentrepreneurs at the individual level, linking MEI registrant to detailed information on their formal employment history and subsequent firm outcomes. We use these data descriptively to document transitions between formal employment and formal microentrepreneurship, as well as the longer-run trajectories of MEI firms after registration. Although descriptive, this analysis provides a valuable complement to the household survey evidence above and helps validate its conclusions.

Transitions Between Formal Employment and Formal Microentrepreneurship: One concern frequently discussed in the Brazilian Policy debate, which is that workers leave formal employment ties to become MEI firm owners because of its tax benefits while still doing the same job for the firm that used to hire them through an employment tie (Alvarez, 2024). This behavior could potentially not show up in the occupation shares of PNAD because the individual might still declare herself an employee, although her formal link is through a MEI firm.[25]

To investigate this possibility, we track transitions between formal employment and formal microentrepreneurship. Our data allows us to individually match microentrepreneurs to their formal employment before and after registering their firm. We build a monthly panel that tracks formal labor market information of all microentrepreneurs for whom we have Social Security Numbers. In each period, we observe if the individual was employed in the formal sector and was a registered microentrepreneur.

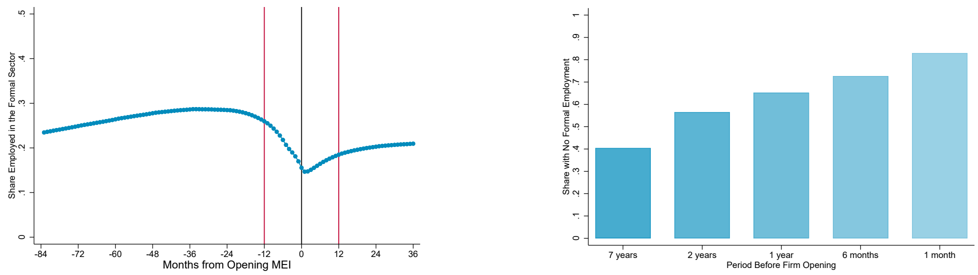
We see in Figure 5a that transitions from formal employment to formal microentrepreneurship are smooth around the period of registration of the firm. To construct it, we pooled all MEI owners for whom we have the Social Security Number and tracked if they worked as formal employees or not up to 7 years before and three years after they registered as MEI firm owners. Around one year before the firm's registration, the share of individuals that are employed starts to decrease from a relatively stable level of 28%. We see that 15% of individuals register as MEI firm owners while holding a formal employment tie. This decrease shows transitions from form, but it also shows that the large majority of MEI firm owners are not transitioning from employment to microentrepeneuship, rather going from an "informal" state where we do not observe them in administrative records, to formalization through the MEI program. Figure 5a also indicates that individuals gaming the labor tax system is not driving the registration of workers as MEI. If individuals were avoiding labor taxes to become MEI, we would expect a sharp decrease around the period they register as MEI, indicating that they were dropping their formal employment tie to register a firm as MEI.[26]

In Figure 5b, we show the share of microentrepreneurs that had no formal tie in the period before opening their firm. We find that 40% of the MEI firm owners did not have a single formal labor market contract in the 7 years before opening their firms. This number is increasing when we get closer to the MEI opening. For 70% of microentrepreneurs, their MEI firm was their only first formal labor market tie in a span of a year.

[25] If all formal microentrepreneurship increase was driven by individuals that would have been formal through employment, then the MEI program would not have generated a reduction in informality, just a reallocation of formal workers to formal microentrepreneurship.

[26] In Figure A19 we present the same exercise but dividing the sample by the year of entry of the MEI firm. This allows us to see the labor market behavior for more years after the MEI firm was opened. We find that MEI firm owners become slightly more likely to transition from the formal labor market to formal microentrepreneurship in more recent cohorts. The behavior after registering as MEI is similar across cohorts.

Figure 5: Share of Microentrepreneurs that were employed in the formal Sector



(a) Share Working **(b)** % No Formal Work Before Opening MEI

Panel A plots the share of individuals that were employed in the formal sector relative to the month at which they opened their MEI. Panel B plots the share of individuals that had no formal labor market tie in the periods before opening their firms. The sample comprises all microentrepreneurs with a valid Social Security Number in CNPJ that registered between Feb. 2010 and Dec. 2015. Being employed in the formal sector means having an active contract registered at RAIS.

Following microentrepreneurs' firms through time: In addition to the transitions between formal employment and MEI firms, our administrative records allow us to follow these individuals' entrepreneurial activities after they register their MEI firm. We combine different administrative sources, allowing us to follow not only the MEI firm, but also other firms created by the same individual.[27]

We find that most microentrepreneurs who register as MEI firms do not grow after they register. Two years after registering as MEI, only 3% of microentrepreneurs are hiring at least one worker. Four years after the firm's registration, only the 99th percentile of active microentrepreneurs hire more than one worker (2 workers). Even among those microentrepreneurs who hire someone, over 70% of them are still hiring only one employee by year 4.[28] These results corroborate our findings from the Household Survey data, that the MEI program had no effects on earnings or firm size in household data, and highlight the lack of growth of newly formalized microentrepreneurs.

5 Tax Revenues, Social Security Access and Fiscal Consequences

Informality is often linked to limited tax collection capacity in developing countries. It is therefore essential to assess whether a program that reduces taxes to promote formalization can generate net fiscal benefits.

Understanding this trade-off requires analyzing the tax revenues and costs of formalization. As shown, commonly cited costs—such as firms opting for the simplified tax regime over the

[27]The full data construction procedures are described in Appendix A.4

[28]We show in Figures A20 and A21 the distribution of firm size and wage bill of firms from microentrepreneurs that registered as MEI.

regular one—do not materialize in our context. However, a significant fiscal cost arises from the contributory benefits of social security that newly formalized individuals become eligible for.

We begin by documenting the revenue gains under the MEI tax system, then estimate the program's effects on contributory social security benefits. Finally, we calculate the overall fiscal impact on the government and conclude with a discussion of the broader implications.

5.1 Tax Revenue Gains

We quantify the increases in government revenue caused by the program, one of the commonly advocated reasons to reduce informality. In the previous sections, we showed that the program drove microentrepreneurs to the formal sector, implying a higher number of formal firms. At the same time, to induce formalization, the program substantially reduced tax rates. Furthermore, there is not necessarily full compliance with paying taxes, which can further reduce the gains in revenue for the government and generate challenges in observing the effects of the program on tax revenues.

To overcome this, we use data from the *Anuário Estatístico da Previdência Social* (AEPS).[29] Yearly, the *Instituto Nacional de Seguridade Social* (INSS), the national autarchy responsible for social security in Brazil, reports its revenues and the number of contributors to the MEI program. This allows us to observe the total value contributed by individuals in the program and also to estimate their compliance with contributions. Given that we find no substitution patterns of individuals shifting to the MEI program from other formal activities, these numbers provide estimates of the causal effect of the program on revenues. We summarize these statistics in Table 3.

The MEI program did not represent significant revenue gains relative to the Brazilian government collection. By 2015, the MEI program represented a yearly revenue of almost 800 Million Reais to the government. This is around 0.1% of government revenues and 0.3% of total social security contributions.[30]

We also find that a large share of registered microentrepreneurs do not comply with their contributions. In the period between 2010 and 2015, the share of defaulters lie around 40-45% of the full sample.[31] Low compliance, however, does not explain why the MEI program does not have meaningful impacts on government revenues, as a simple calculation indicates that even under full compliance, its revenues would represent a very small share of both total Government and Social Security collections.

[29] See Appendix A.3 for a description of the AEPS data.

[30] Total government revenue in 2015 was 820 Billion Reais and Social Security Contributions amounted to 240 Billion Reais (in 2009 Values).

[31] These numbers are consistent to the findings from (Bosch Mossi et al., 2015), who estimate an average default rate of 41.6% for MEI affiliates between 2011-2015 using similar administrative data from the Ministry of Social Security.

Table 3: Descriptive statistics of the MEI contributions to social security

Year	Number of MEI Firms	Number of MEI Contributors	$ MEI Contributions (BRL 2009)	Percentage of MEI Defaulters
2010	949,720	557,540	167,402,668	41%
2011	1,844,149	995,289	243,531,412	46%
2012	2,854,975	1,523,131	335,067,859	47%
2013	3,884,171	2,164,620	522,061,304	44%
2014	4,928,719	2,854,889	686,130,839	42%
2015	6,019,331	3,395,337	795,306,794	44%

Notes: This table presents descriptive statistics of the MEI contributions to the Brazilian pension system. The first column following the year brings counts of the CNPJ data as a reference. The remaining information comes from the *Anuário Estatístico da Previdência Social*/INSS.

5.2 Social Security Coverage

Next, we quantify the effects of the MEI program on contributory social security benefits. As in many countries (Levy, 2008), access to social security in Brazil is largely conditional on formal workforce participation. Thus, by driving microentrepreneurs out of informality, the MEI program also started covering them in the social security system.

In what follows, we use data that covers the universe of social security benefits paid by the INSS. This is a public data set organized at the benefit level, which includes the start and end dates, geographic location, benefit category, and the class of contribution associated with each benefit. Benefits are not identified at the individual level; thus, we cannot directly map the probability that a given individual receives a benefit. In turn, we can look at aggregates by *class*, which allows us to observe how many benefits were issued to self-employed individuals, relative to the other groups that are eligible to INSS coverage. This allows us to quantify the role of the MEI program in the aggregate number of benefits provided by INSS - the relevant metric for fiscal consequences. Furthermore, we disentangle benefits by category (e.g., maternity, health, retirement, pensions) to discuss the implications of changes in coverage.[32]

Our analysis consists of a difference in differences model that compares how benefits evolved before and after the implementation of the MEI program for different classes of coverage. We consider the Individual Contributors class to be our treated group. That encompasses entrepreneurs and self-employed individuals. We use all the remaining affiliate types in the control group. That includes wage-dependent workers, special insured, and facultative affiliates.[33]

We then aggregate our data at a quarter x state x class level and estimate the following equation:

[32] See Appendix A.3 for a detailed description of the INSS benefits data.

[33] Refer to appendix A.3 for a complete description of the INSS affiliate classes.

$$Y_{out} = \alpha_o + \delta_{ut} + \sum_{t=2005}^{t=2015} \beta_t \cdot D_{ot} + \varepsilon_{out} \tag{4}$$

where α_o, δ_{ut} are fixed effects by State times quarter, and α_o is an indicator for the class. D_{ot} is an indicator that the class is treated (Individual Contributors), and the period is equal to t.

We focus our analysis on two primary outcomes: the log of benefits issued and the log of total benefits that are active at a given moment. The identification assumption is that in the absence of the program, the trends of our outcomes would be parallel across different classes of individuals covered by social security.

We estimate that the MEI program substantially increased the number of social security benefits issued to individual contributors and the stock of active benefits. Our estimates suggest that by 2015, there were around 55% more benefits issued to the class of individual contributors. This reflects in around 15% more benefits being paid to this class by the end of 2015. Our estimates are shown in Figures 6a and 6b.

We highlight two important patterns in our results that corroborate to the causal interpretation of our findings. First, we see the absence of any differential trends in our coefficients before the program implementation, indicating the validity of our identification assumption. Second, the effects are delayed relative to the start of the program. This is reassuring as MEI firm owners have to contribute to Social Security for at least 10 months to be covered by the majority of benefits from INSS. This also lines up with our findings on the number of MEI contributors, which is increasing over the period.[34]

5.3 Costs of the program to the Government

With our findings, we can calculate what are the net government costs of the program. We calculate it as the changes in expenditures and the gains in revenue caused by it:

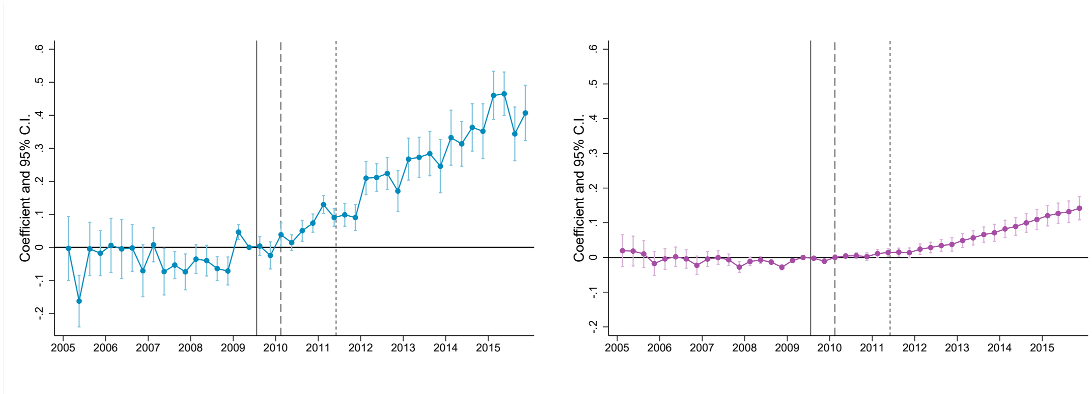
$$\text{Net Gov. Cost} = \Delta E - \Delta R$$

where ΔE denotes the government expenditures with the policy, and ΔR the gains in government revenue caused by it.

We use the revenue collected from contributions of MEI firm owners as our measure of gains in government revenue (ΔR). Given our findings that (i) the program did not shift firms from the regular tax systems to the MEI tax system, (ii) microentrepreneurs that registered as MEI would have been informal in the absence of the program, it is reasonable to assume that in the absence of the MEI program, the revenue from contributions would not have been collected.

[34] The difference between our findings on issued and active benefits can be explained by the effects of the MEI program being concentrated on shorter-term benefits. We replicate our exercise, decomposing our outcome by type of benefit in Table A7. There, we see that longer-term benefits such as retirement are not affected by the program. We discuss these findings below.

Figure 6: Effects on the Social Security Benefits



(a) Log of Benefits Issued

(b) Log of Active Benefits

This Figures show estimates β_t from equation 4. Panel (a) uses as the outcome the log of benefits issued, whereas Panel (b) shows the log of benefits that are active in the given quarter. The benefit-level data is aggregated at the State x Quarter x Class level. Vertical lines indicate the start of the different phases of the program. Standard Errors are clustered at the State and Class levels.

Thus, the changes in revenues caused by the policy are equivalent to what was collected by MEI contributors.

The main expenditures with the program come from social security benefits given to formalized individuals. Using our findings from the previous section, we can calculate how much this represented in terms of government funds. To do so, we estimate the predicted benefits handed to the treatment group in the absence and the presence of the policy in each state and each period, $\hat{B}^0_{ut}, \hat{B}^1_{ut}$ respectively.

$$\hat{B}^0_{ut} = \hat{\alpha}_o + \hat{\delta}_{ut} \tag{5}$$

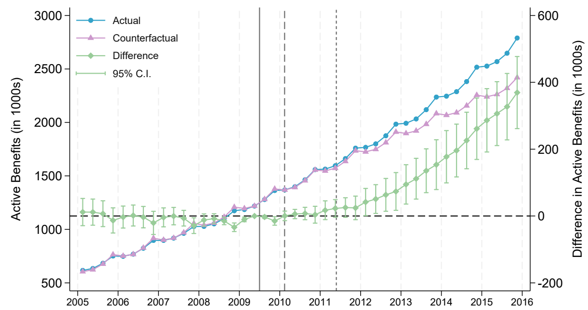
$$\hat{B}^1_{ut} = \hat{\alpha}_o + \hat{\delta}_{ut} + \hat{\beta}_t \cdot D_{ot} \tag{6}$$

where $\{\hat{\alpha}_o, \hat{\delta}_{ut}, \hat{\beta}_t\}$ are the coefficients estimated in equation 4.

We then compute the exponential of $\hat{B}^0_{ut}, \hat{B}^1_{ut}$ and sum the total number of benefits across states in each period. This gives us estimates of the actual number of benefits in a given period and the counterfactual in the absence of the program. We plot these estimates together with the difference between them in Figure 7.

Our estimates indicate that in 2015, social security issued, on average, 322,280 additional benefits to Individual Contributors due to the MEI program (between 223,190 and 421,368 with 95% confidence). We translate this increase into fiscal terms by multiplying the number of additional benefits by the monthly minimum wage of BRL 527 (BRL 788 in 2009 Brazilian Reais values) and annualizing the resulting amount over 12 months. This implies an increase in annual government expenditures of BRL 2.038 billion, with a corresponding 95% confidence

Figure 7: Estimates of Total Benefits Active to Individual Contributors



Notes: This figure plots our estimates of the actual and counterfactual benefits for Individual Contributors in the Social Security data. In the right y-axis we plot the difference between the two measures along with the 95% confidence intervals, which we compute using the delta method.

interval ranging from BRL 1.411 billion to BRL 2.664 billion.

Accounting for program revenues of BRL 0.795 billion in 2015, our estimates imply that the program's net fiscal cost in 2015 ranged from BRL 0.62 billion to BRL 1.87 billion (95% confidence interval), with our preferred estimate equal to BRL 1.243 billion, derived from the point estimate of the annual expenditure increase:

$$\text{Net Gov. Cost in 2015 (Billions)} = \Delta E - \Delta R = 2.038 - 0.795 = 1.243.$$

This corresponds to a net government cost of, approximately, BRL 207 per registered microentrepreneur per year. In relative terms, our findings imply that expenditures on social security are about 2.56 times higher than the tax revenues collected by the government (between 1.77 and 3.35 with 95% confidence).

The fiscal effects can also be assessed by examining a counterfactual in which the monthly tax rate remained at 11%, using a simple back-of-the-envelope calculation. Assuming the number of benefits and overall compliance remained unchanged—a plausible scenario given that we find no significant differential effect of the tax cut on firm registration and no clear shift in compliance after the reform—we estimate that revenues would have been 2.2 times higher. This would correspond to 1.749 billion BRL in revenues in 2015, implying that the program costs would have exceeded revenues by only 16%.

5.4 MVPF and Discussion

Our findings above indicate that the targeted tax system was not effective in improving the government's fiscal capacity. However, the simple government accounting exercise ignores the benefits of social security coverage to the microentrepreneur population. This speaks directly to a longstanding challenge in Brazil and Latin America in general: the difficulty of expanding contributory social security coverage in economies where a large share of workers are self-employed or move frequently between formal and informal jobs. Because contributory benefits depend on stable formal employment and regular payroll contributions, many workers accumulate low contribution densities and remain effectively outside the system (Levy and Schady, 2013).

To illustrate this, it is useful to understand what types of benefits are being issued to microentrepreneurs. In Table A7, we present estimates of our difference in differences analysis, decomposing the effects on the aggregate number of benefits issued by types of benefits. We aggregate the INSS contributory benefits in four main categories. First, short-term contributory benefits, including sickness benefit (Auxílio-Doença), work-accident benefit (Auxílio-Acidente), or incarceration benefit (Auxílio-Reclusão). Second, maternity benefits. Third, survivor pensions, including both standard contributory pensions and those related to occupational accidents. Fourth, retirement benefits, encompassing retirement by age, length of contribution, or disability.

The main increase occurs in maternity benefits. We also observe small increases in health-related and retirement benefits, but no detectable effects on disability or old-age pensions. The pattern for maternity benefits is fully consistent with the heterogeneity results documented earlier: women experience substantially larger increases in formalization than men, to the point that the pre-existing gender gap in formalization among microentrepreneurs essentially disappears.

Such benefits are associated with substantial positive effects for beneficiaries. Maternity benefits, for example, have been shown to improve children's short-term outcomes (Rossin, 2011) and long-term human capital (Carneiro et al., 2015). In Brazil, GC Britto et al. (2024) find that maternity benefits issued to unemployed women can reduce criminal activity among first-time fathers. There is also substantial evidence that health-related fiscal benefits mitigate adverse effects of shocks on household members (Fadlon and Nielsen, 2019; Coile et al., 2022).

A more objective way to evaluate these expenditures is through the Marginal Value of Public Funds (MVPF) framework (Hendren and Sprung-Keyser, 2020), which divides individuals' willingness to pay for the program by its net costs to generate a unified, comparable measure of the value of public spending. To calculate the numerator of the MVPF, we assume that individuals' willingness to pay for the benefits they receive is one-to-one with the amount received. Given a willingness to pay of one for each BRL spent on benefits[35], and the net cost of the program described above, we calculate the MVPF of the MEI program to be 1.64. $\text{MVPF} = \frac{\text{Willingness to Pay}}{\text{Net Costs}} = \frac{2.038}{1.243} = 1.64$

Our assumptions in building the MVPF are relatively conservative, as they do not take into account the positive spillovers of social security spending discussed above. These spillovers can potentially increase the numerator, as these benefits may have positive effects on other household members' long-term outcomes. They could also potentially reduce the denominator if

[35] This assumption is the same as that of Hendren and Sprung-Keyser (2020) when analyzing the MVPF of disability insurance in Autor et al. (2016) and Gelber et al. (2017), and by Paradisi (2021) when computing the MVPF of maternity leave using estimates from Denmark in Brenøe et al. (2024).

benefits reduce government expenditures on health care, for example. Despite this conservative approach, our MVPF estimate is relatively high compared to existing studies on taxes or social insurance,[36] suggesting an adequate social return relative to fiscal cost. This reflects the fact that MEI owners largely transition from informality into the tax base, thereby expanding government revenues, rather than reallocating firms from the other tax regimes.

At the same time, our framework abstracts from potential long-run retirement pension liabilities that may arise if newly formalized workers accumulate sufficient contribution histories to claim contributory old-age benefits. Incorporating these future obligations would mechanically increase the program's net fiscal cost and lower the estimated MVPF. However, this concern is mitigated by the relevant counterfactual. Many of the individuals induced to formalize under the MEI program would likely qualify for non-contributory old-age assistance, such as the Benefício de Prestação Continuada (BPC), even in the absence of formalization. As a result, formalization primarily shifts beneficiaries from non-contributory to contributory old-age programs, rather than generating entirely new old-age expenditures. To the extent that contributory pensions substitute for BPC transfers later in life, the net impact of the program on age-based retirement expenditures is therefore likely to be limited.

Overall, our analysis suggests that targeted tax systems designed to promote formalization are unlikely to increase government revenues directly. However, when the fiscal costs primarily take the form of transfers through social insurance, such policies can still generate a social return relative to their fiscal cost.

6 Conclusion

Informality and microentrepreneurship continue to shape labor markets across developing countries. Most businesses in these settings are small and operate informally, creating a series of well-known challenges: reduced tax revenues, limited access to social security, and the persistent association between informality and lower productivity and earnings. These concerns have motivated policymakers to explore interventions aimed at increasing formalization, particularly by reducing the monetary and administrative costs of becoming formal.

In this context, the Brazilian MEI program offers unique evidence on how a highly targeted tax system can reshape formalization decisions among microentrepreneurs. By substantially reducing both the monetary and administrative burdens of becoming formal, the program directly addressed the barriers faced by a large and vulnerable segment of the workforce.

Our findings show that the targeted tax regime meaningfully lowered the costs of formalization and, in doing so, effectively shifted a large share of microentrepreneurs from informality into the formal sector. While the reform did not translate into higher fiscal capacity or sizable productivity improvements among those who formalized—objectives often associated with

[36] The median MVPF in 33 tax-related studies in Policy Impacts is 1.3, and among 61 studies related to Social Security Policies is 1.026.

formalization programs—it did bring a substantial number of workers into the social security system. This expansion of coverage represents an important welfare gain for a particularly vulnerable segment of the labor force, underscoring the broader social relevance of the policy even in the absence of positive government net revenue or improvements in firm performance.

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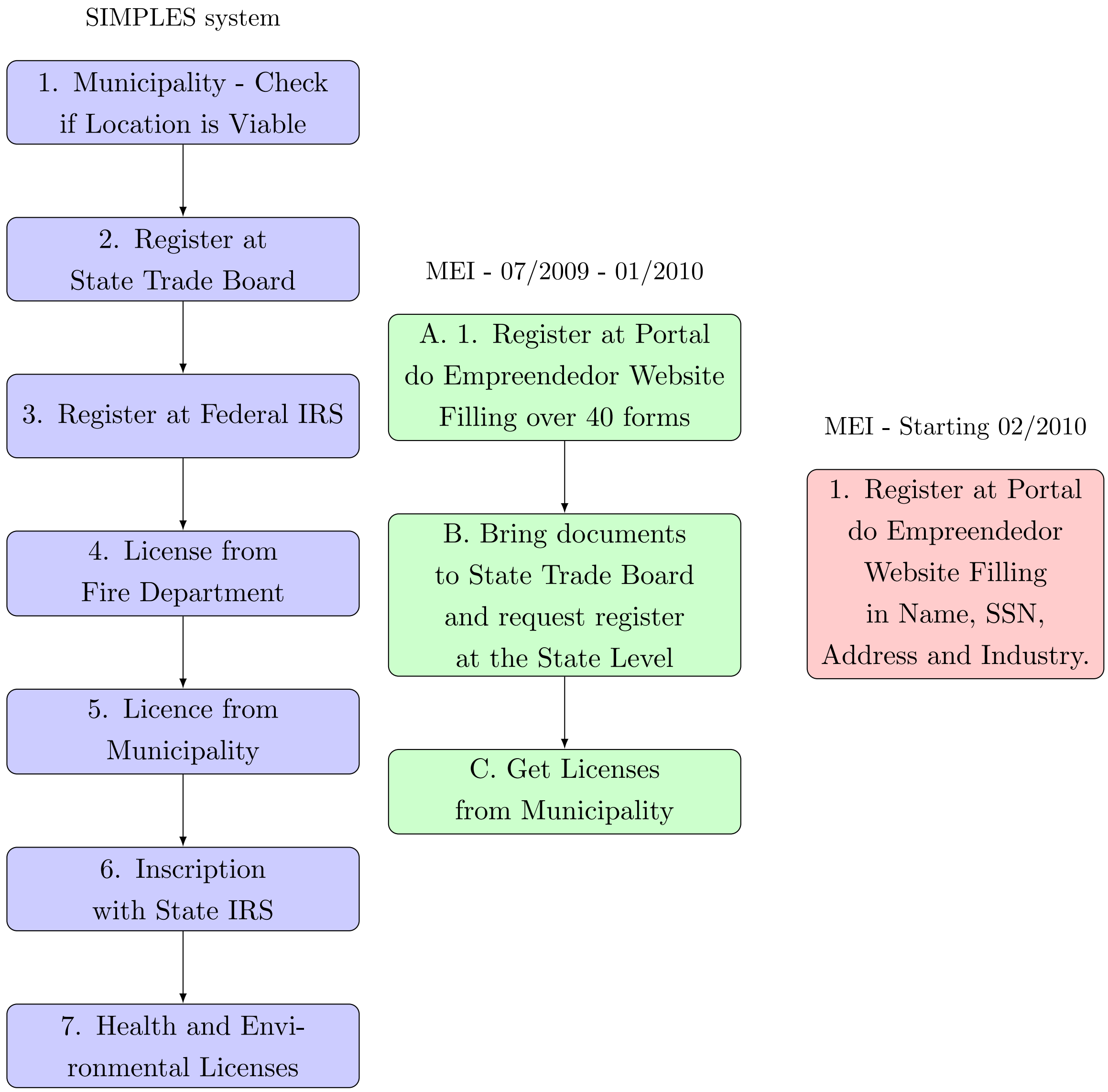
Declaration of generative AI and AI-assisted technologies in the writing process:

During the preparation of this work the authors used ChatGPT 4.0 and 5.2 in order to review grammatical errors, improve language and readability. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Table of Contents - Appendix

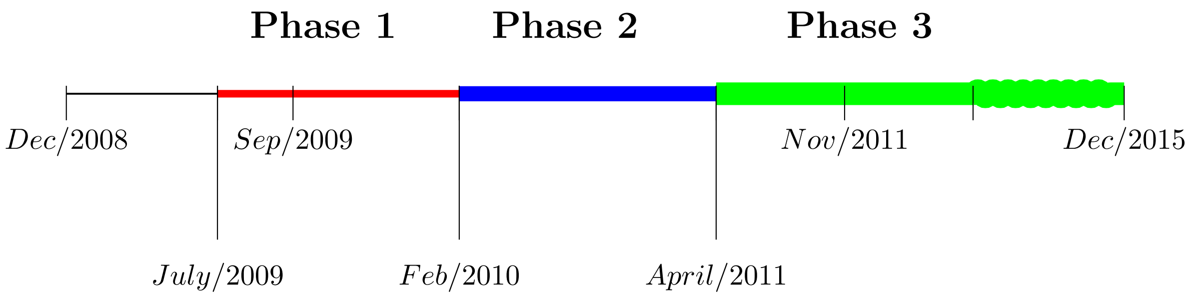
Additional Figures

Figure A1: Steps Necessary to Register a Firm



This figure summarizes the process to register a firm in the SIMPLES system, and in the MEI system, before and after the creation of the integrated system in February 2010.

Figure A2: Timeline of The MEI program



Event Details:

- **December 2008:** The project was sanctioned by the federal government.

- **July 2009: Beginning of Phase 1 (Implementation Phase)** Implementation began in a selected group of states (SP, RJ, DF, MG), and the integrated website is not ready yet.

- **September 2009:** another group of states were allowed (CE, ES, PR, RS, SC), integrated website was not ready yet.

- **February 2010 Beginning of Phase 2:** All states included and integrated website started to work.

- **April 2011 Beginning of Phase 3:** Tax rate reduced from 11% of the Minimum Wage to 5%.

- **November 2011:** Eligibility revenue cap raised to 60,000 R$.

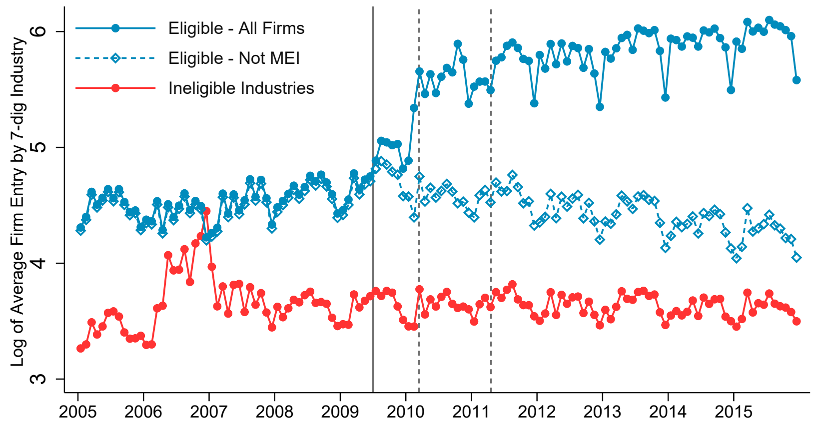
Short Summary of Formalization Costs at each Phase:

- **Phase 1:** Tax rate at 11% of Minimum Wage. Registration costs are lowered, but the integrated website is not working and is still costly to register firms.

- **Phase 2:** Tax rate still at 11% of Minimum Wage. Registration costs were virtually eliminated, and the program worked throughout the whole country.

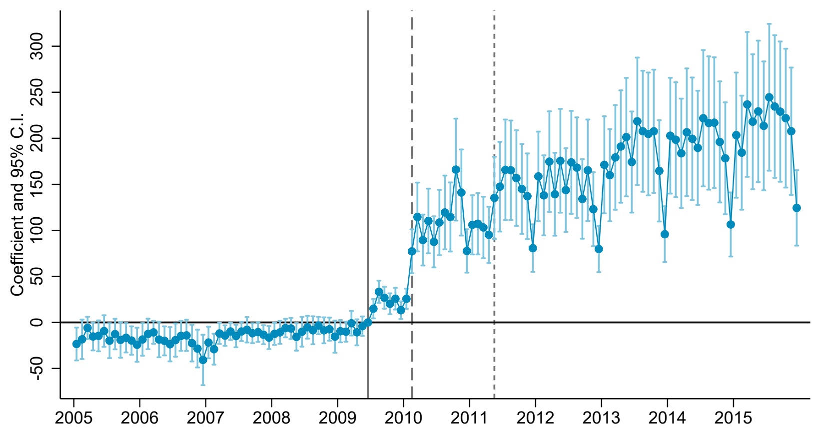
- **Phase 3:** Tax rate at 5% of Minimum Wage. Registration costs were virtually eliminated, and the program worked throughout the whole country.

Figure A3: Average Log of Firm Registration By Industry Eligibility



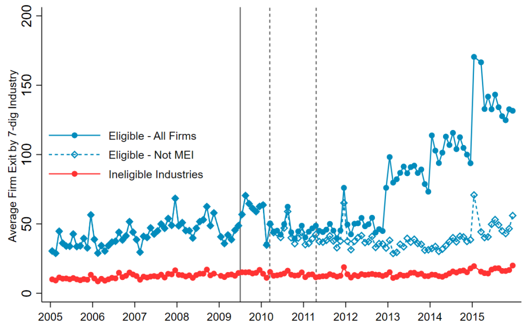
Notes: This figure plots the log of average firm registration by industry for eligible and ineligible 7 digit CNAE classification. List of eligible industries were defined from the *CGSN 58* at the 7 digit CNAE. The blue connected line with solid markers shows the average firm registration in eligible industries for firms in all types of tax systems (MEI and regular firms). The blue connected line with hollow markers shows the average firm registration of firms not in the MEI program for industries in the eligible sectors. The red line shows average firm registration in ineligible industries. Vertical lines represent the start of the different phases of the program.

Figure A4: Difference in Differences Coefficients on Firm Entry - Levels

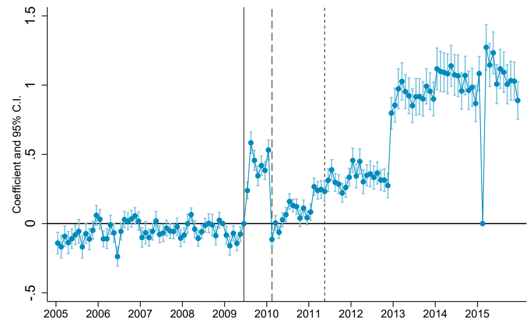


This figure plots the vector of coefficients β_t and 95% Confidence Intervals from equation 1. The sample consists of a monthly panel at the 7-digit industry level from 2005 to 2015. The outcome is the number of firms entering each period. Vertical lines represent the start of the different phases of the program. Standard errors are clustered at the 7-digit CNAE level.

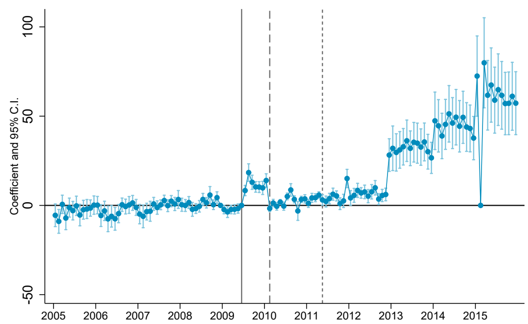
Figure A5: Effects on Firm Exit



(a) Average by Industry



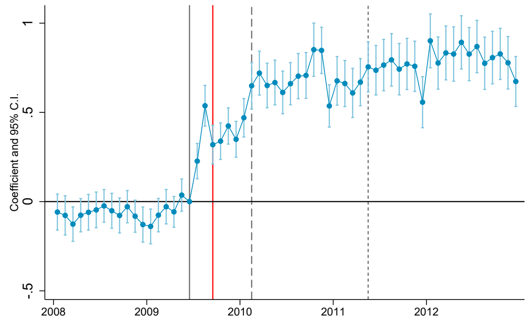
(b) Estimates on Log of Firm Exit



(c) Estimates on Levels of Firm Exit

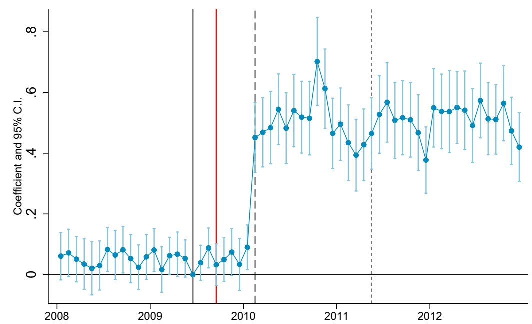
Notes: Panel A plots the average firm exit by industry for eligible and ineligible 7 digit CNAE classification. List of eligible industries were defined from the *CGSN 58* at the 7 digit CNAE. The blue connected line with solid markers shows the average firm exit in eligible industries for firms in all types of tax systems (MEI and regular firms). The blue connected line with hollow markers shows the average firm exit of firms not in the MEI program for industries in the eligible sectors. The red line shows average firm exit in ineligible industries. Vertical lines represent the start of the different phases of the program. Panels B and C plots the estimates of equation 1 on both logs and levels of firm exit.

Figure A6: Difference in Differences Coefficients on Log of Firm Entry - by State



(a) SP, RJ, MG, DF

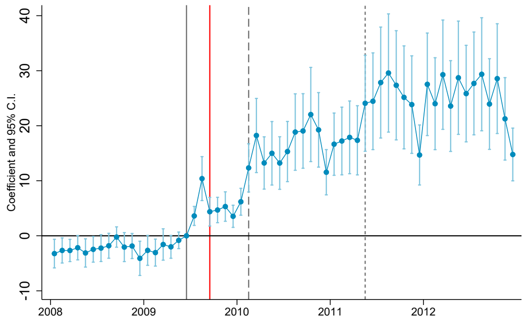
(b) RS, SC, PR, CE



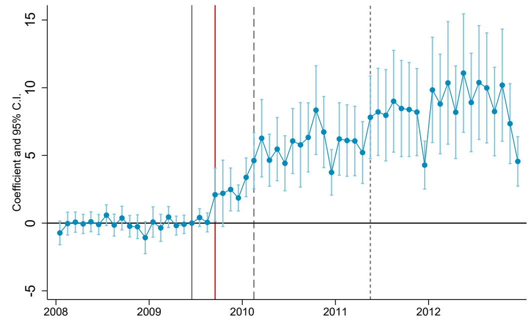
(c) Remainder States

This figure plots the vector of coefficients β_t and 95% Confidence Intervals from equation 1. The sample consists of a monthly panel at the 7-digit industry level from 2008 to 2012. The outcome is the log of the number of firms entering each period. Vertical gray lines represent the start of the different phases of the program. Vertical red line represents September 2009 Standard errors are clustered at the 7-digit CNAE level. SP, RJ, MG and DF were allowed into the program in July 2009, RS, SC, PR and CE were allowed in September 2009, and remainder states were allowed in February 2010.

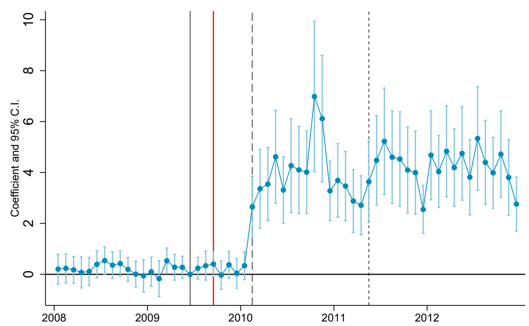
Figure A7: Difference in Differences Coefficients on Firm Entry - Levels by State



(a) SP, RJ, MG, DF



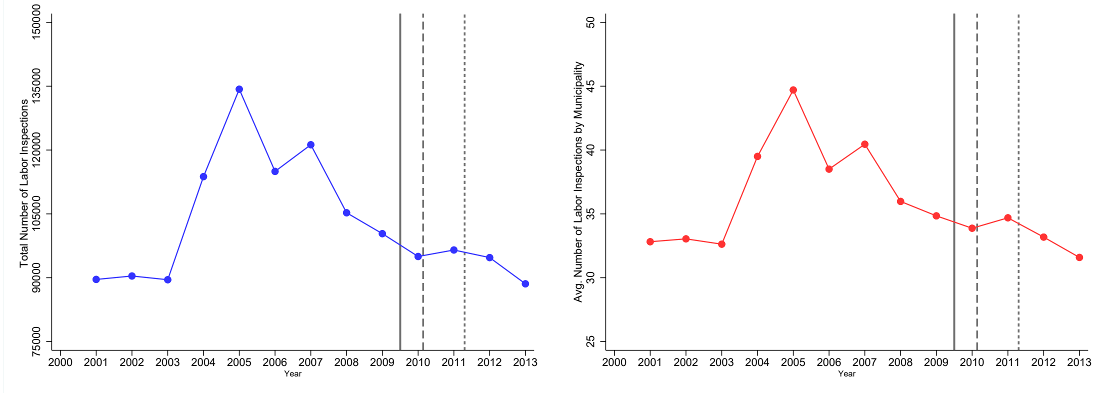
(b) RS, SC, PR, CE



(c) Remainder States

This figure plots the vector of coefficients β_t and 95% Confidence Intervals from equation 1. The sample consists of a monthly panel at the 7-digit industry level from 2008 to 2012. The outcome is the number of firms entering each period. Vertical gray lines represent the start of the different phases of the program. Vertical red line represents September 2009 Standard errors are clustered at the 7-digit CNAE level. SP, RJ, MG and DF were allowed into the program in July 2009, RS, SC, PR and CE were allowed in September 2009, and remainder states were allowed in February 2010.

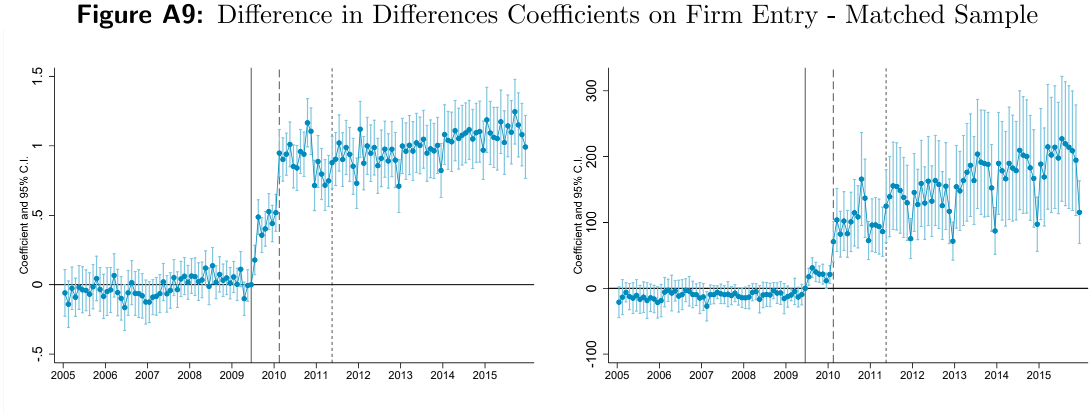
Figure A8: Inspections by Year



(a) Total Inspections in the Country **(b)** Avg. Inspections by Municipality

This figure plots trends in the number of firms inspected by the Ministry of Labor and Employment (MTE) between 2000 and 2013. Data on inspections was collected from the replication package of (Ponczek and Ulyssea, 2022). Panel (a) plots the total number of firms inspected in the country, whereas Panel (b) plots the average number of firms by municipality conditional on having at least one firm inspected in each given year in that municipality.

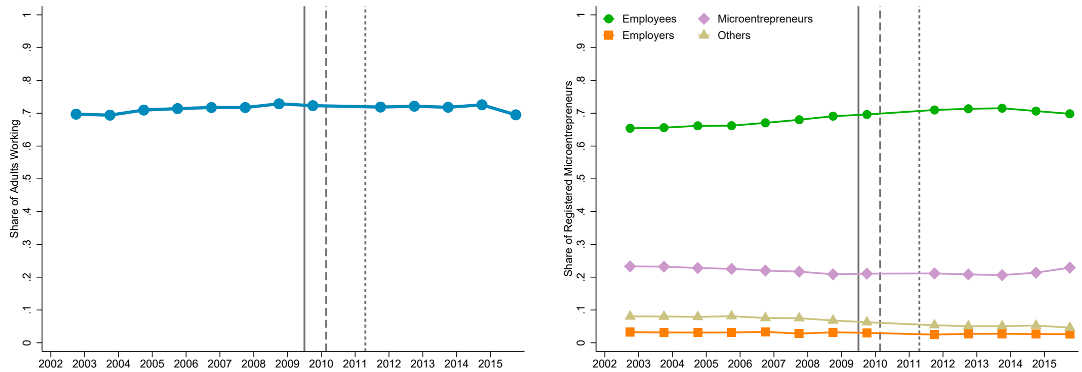
Figure A9: Difference in Differences Coefficients on Firm Entry - Matched Sample



(a) Log of Firm Entry **(b)** Levels of Firm Entry

This figure plots the vector of coefficients β_t and 95% Confidence Intervals from equation 1. The sample consists of a monthly panel at the 7-digit industry level from 2005 to 2015 using industries matched. Match is done without replacement using a propensity score matching that estimates propensity to treatment using 2008 firm entry, number of active firms, and number of firms in the SIMPLES system. The outcome is the log of (Panel A) and the levels (Panel B) of the number of firms entering each period. Vertical lines represent the start of the different phases of the program. Standard errors are clustered at the 7-digit CNAE level.

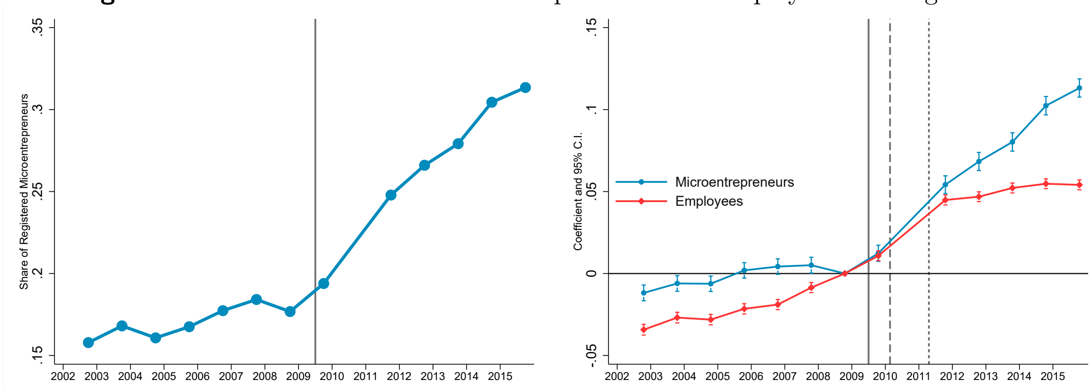
Figure A10: (Lack of) Occupational Change



(a) Share Working **(b)** Share by Occupation Category

Panel A plots the share of individuals between 18 and 60 years that are working. Panel B plots the share of workers between 18 and 60 years in each type of occupation. Microentrepreneurs are defined as individuals that declare themselves self-employed or firm owners with at most one employee. Employers are firm owners that have more than one employee. Employees include formal and informal workers that work for a monetary income in a firm or in the public sector. Others category include workers that work for non monetary benefit which mostly encompasses farmers in that work for their own consumption. Vertical lines represent the start of the different phases of the program. All results were built with available PNAD data from 2002 - 2015. 2010 values are missing because PNAD was not collected in that year.

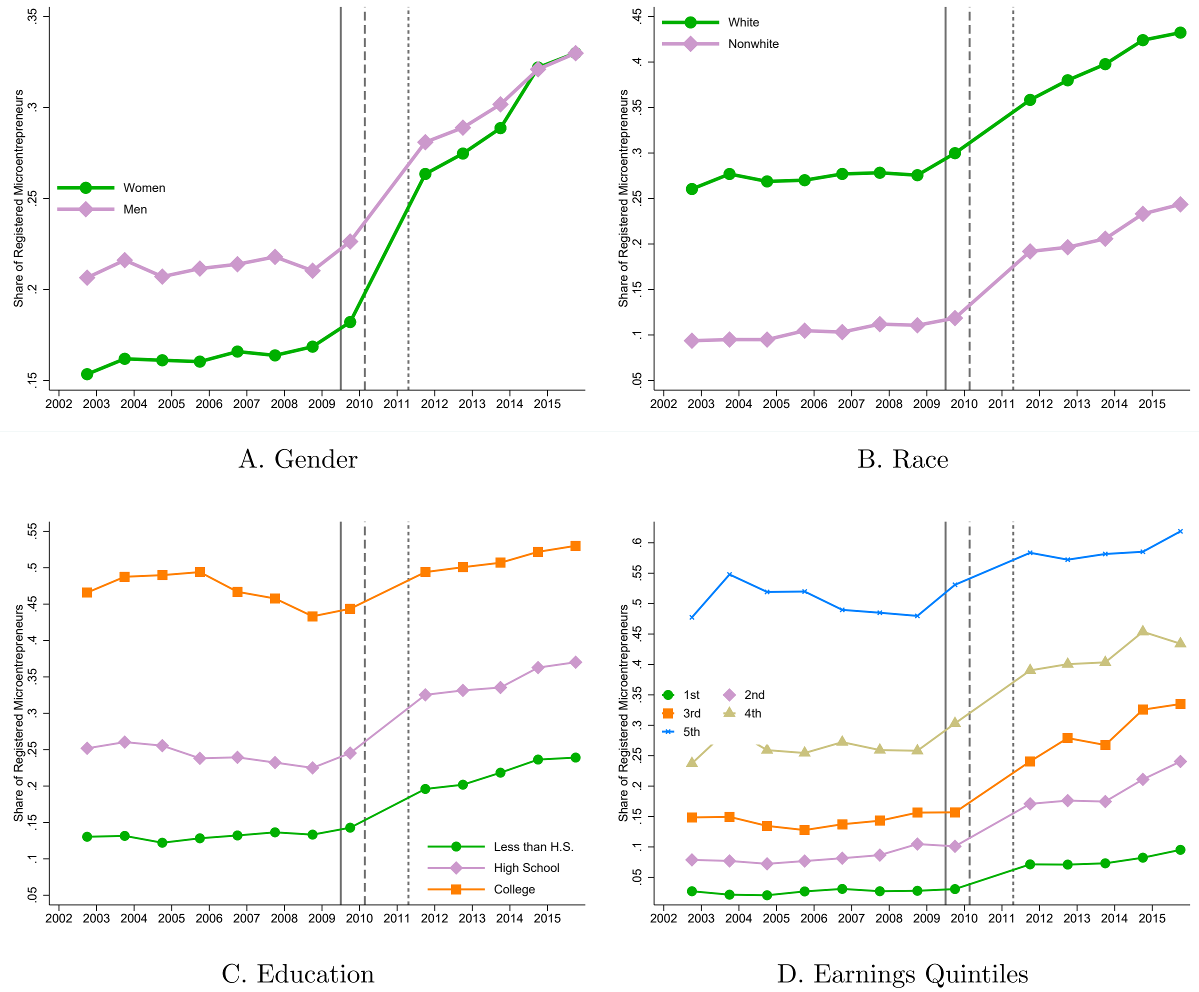
Figure A11: Share of Formal Microentrepreneurs and Employees with Agriculture



(a) Microentrepreneurs **(b)** Controlling by Observables

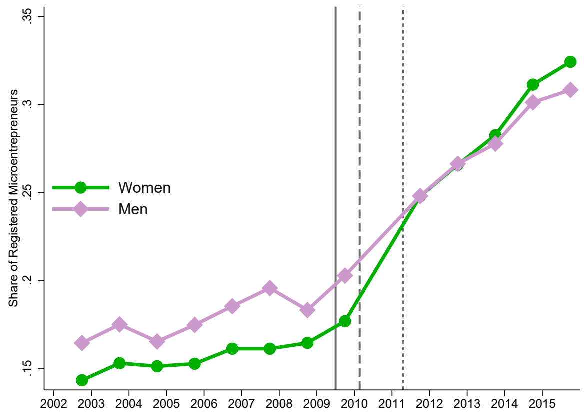
Microentrepreneurs are defined as individuals who declare themselves as self-employed or employers who hire at most one employee. We define a formal microentrepreneur as those who declare to be contributing to social security through their job. Panel A plots the share of microentrepeneurs between 18 and 60 years that are formal by year. Panel B plots the coefficients α_t of two separate regressions. The blue one, is with the sample of microentrepreneurs. The red one with the sample of individuals working as employees of a firm. All results were built with available PNAD data from 2002 - 2015. 2010 values are missing because PNAD was not collected in that year.

Figure A12: Share of Formal Microentrepreneurs by Group

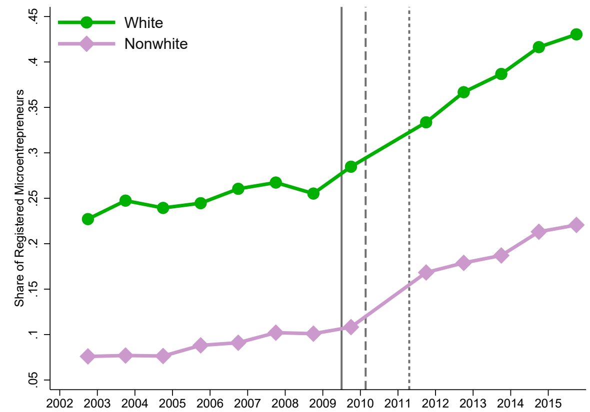


Notes: All results were built with available PNAD data from 2002 - 2015. 2010 values are missing because PNAD was not collected in that year. Sample is restricted to microentrepreneurs, which are defined as individuals that declare themselves self-employed or firm owners with at most one employee. Sample is restricted to individuals between 18 and 60 years. Panel A divides microentrepreneurs in three broad education groups by the highest degree that they completed: Less than High School, High School and More than High School. Panel B divides the sample in earnings quintiles. Panel C plots the share of microentrepreneurs that are formal by gender. Panel D plots the share of microentrepreneurs years that are formal by race. Vertical lines represent the start of the different phases of the program.

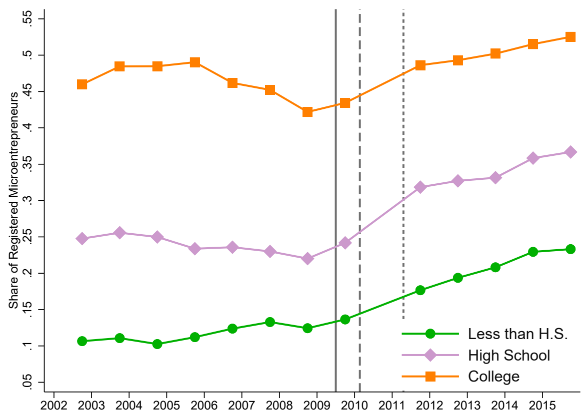
Figure A13: Share of Formal Microentrepreneurs by Group including Agriculture



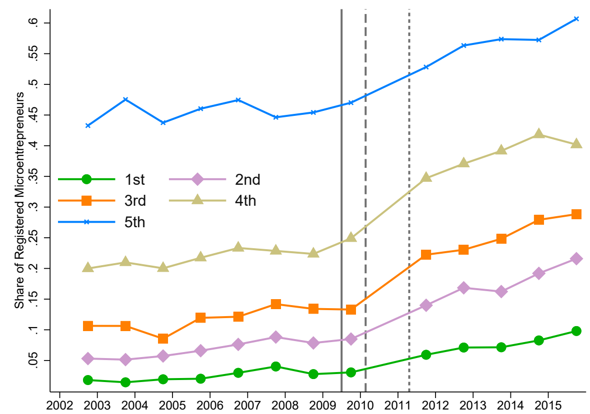
A. Gender



B. Race



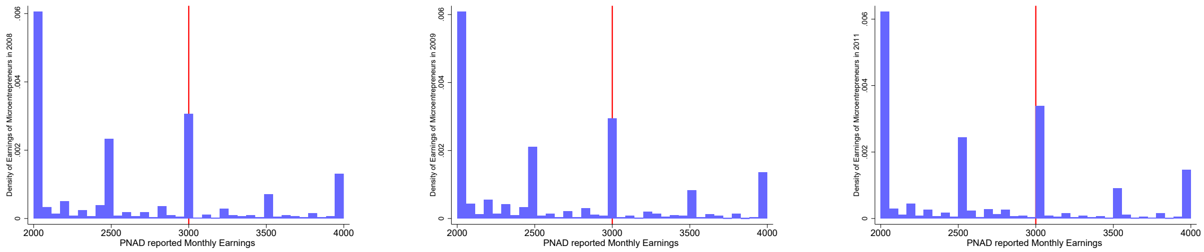
C. Education



D. Earnings Quintiles

Notes: All results were built with available PNAD data from 2002 - 2015. 2010 values are missing because PNAD was not collected in that year. Sample is restricted to microentrepreneurs, which are defined as individuals that declare themselves self-employed or firm owners with at most one employee. Sample is restricted to individuals between 18 and 60 years. Panel A divides microentrepreneurs in three broad education groups by the highest degree that they completed: Less than High School, High School and More than High School. Panel B divides the sample in earnings quintiles. Panel C plots the share of microentrepreneurs that are formal by gender. Panel D plots the share of microentrepreneurs years that are formal by race. Vertical lines represent the start of the different phases of the program.

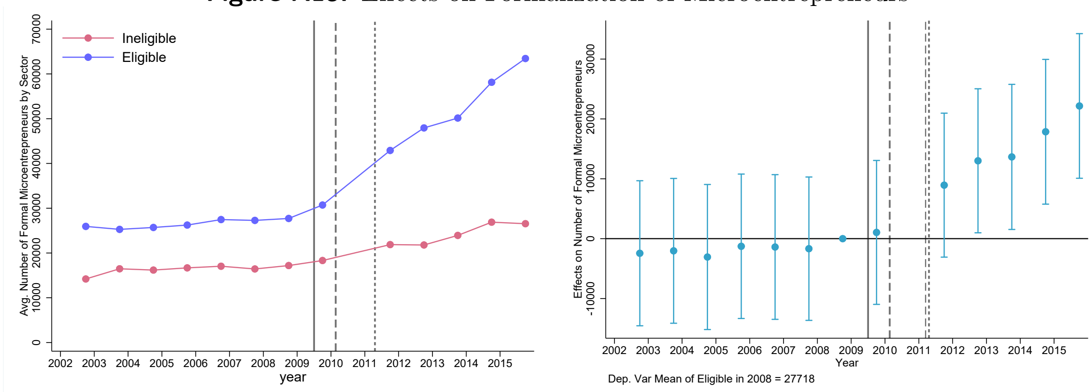
Figure A14: Distribution of Microentrepreneurs' Earnings around Eligibility cutoff in PNAD



2008 2009 2011

This figure shows histograms of the monthly earnings distribution of microentrepreneurs in PNAD 2008-2011. The red line shows the monthly equivalent of the MEI cutoff. Microentrepreneurs are those who declare themselves self-employed or employers with at most one employee.

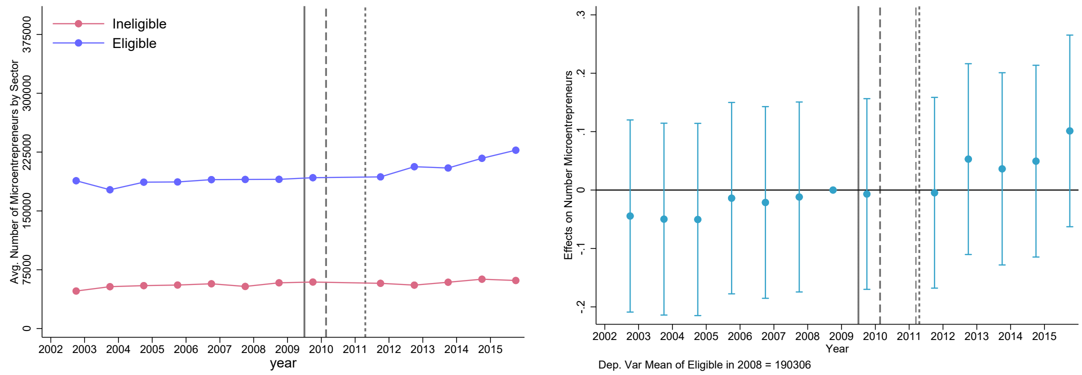
Figure A15: Effects on Formalization of Microentrepreneurs



(a) Formal Microentrepreneurs - Trends **(b)** Formal Microentrepreneurs

This figure shows formalization of microentrepreneurs across eligible and ineligible industries using PNAD data. Panel (a) shows the number of formal microentrepreneurs among eligible and ineligible industries. Panel (b) plots the coefficients β_t from equation 3. The sample comprises all microentrepreneurs in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state. All results were built with available PNAD data from 2002 - 2015. 2010 values are missing because PNAD was not collected in that year.

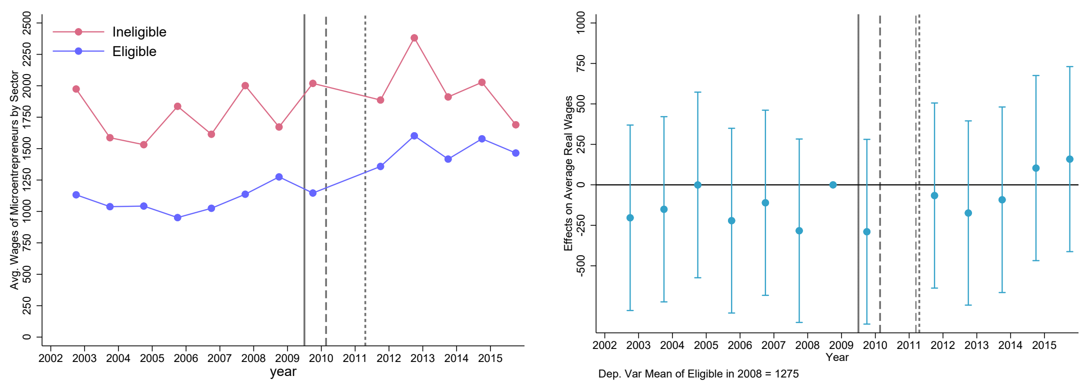
Figure A17: Effects on Number of Microentrepreneurs



(a) Number of Microentrepreneurs - Trends **(b)** Number of Microentrepreneurs

This figure shows the average number of microentrepreneurs across eligible and ineligible industries using PNAD data. Panel (a) shows the number of formal microentrepreneurs among eligible and ineligible industries. Panel (b) plots the coefficients β_t from equation 3 divided by the average of the outcome variable in 2008. The sample comprises all microentrepreneurs in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state. All results were built with available PNAD data from 2002 - 2015. 2010 values are missing because PNAD was not collected in that year.

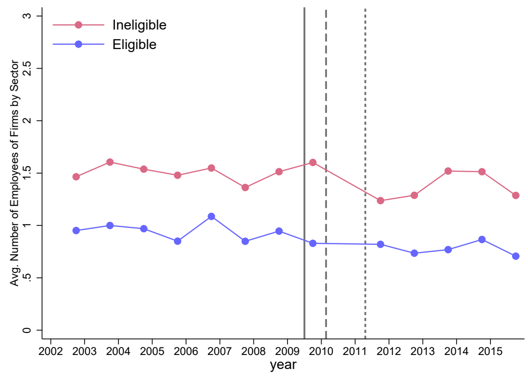
Figure A16: Effects on Labor Earnings of Microentrepreneurs



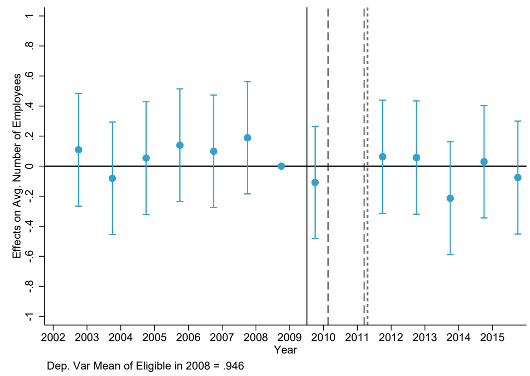
(a) Average Labor Earnings - Trends **(b)** Average Labor Earnings

This figure shows the average labor earnings of microentrepreneurs across eligible and ineligible industries using PNAD data. Labor Earnings are adjusted to December 2018 values. Panel (a) shows the number of formal microentrepreneurs among eligible and ineligible industries. Panel (b) plots the coefficients β_t from equation 3. The sample comprises all microentrepreneurs in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state.

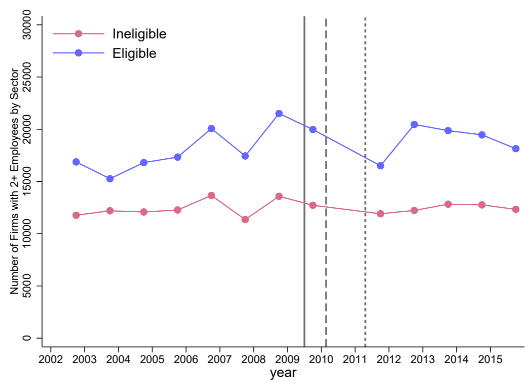
Figure A18: Effects on Firm Size



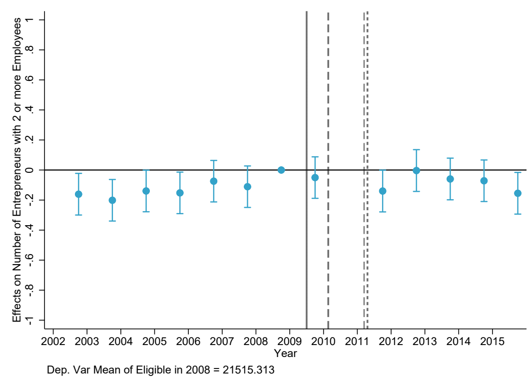
(a) Avg. Number of Employees - Trends



(b) Avg. Number of Employees



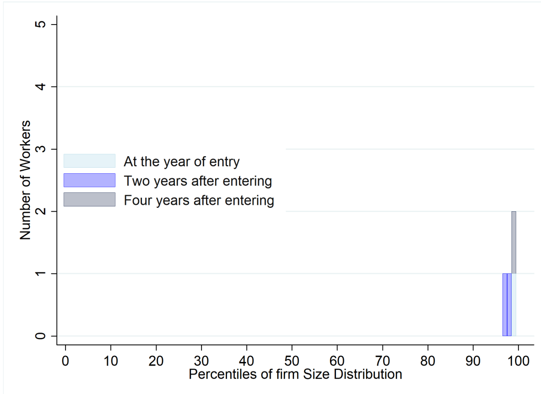
(c) Entrepreneurs with ≥ 2 Workers



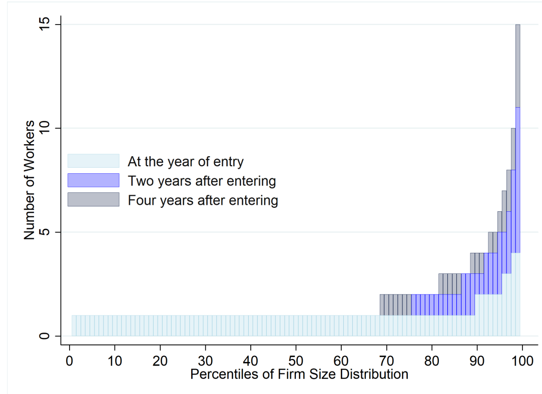
(d) Number of Entrepreneurs with ≥ 2 Workers

This figure shows results on firm size across eligible and ineligible industries using PNAD data. Panel (a) shows the average number of employees of firms in eligible and ineligible industries. Panel (c) shows the number of employers with 2 or more employees in PNAD. Panels (b) and (d) plots the coefficients β_t from equation 3 divided by the average of the outcome variable in 2008. The sample comprises all self employed or employers in between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state. The number of employees is a categorical variable in RAIS. We use the minimum of each category as the input for the number of employees. All results were built with available PNAD data from 2002 - 2015. 2010 values are missing because PNAD was not collected in that year.

Figure A20: Firm Size Distribution of Microentrepreneurs that Register as MEI



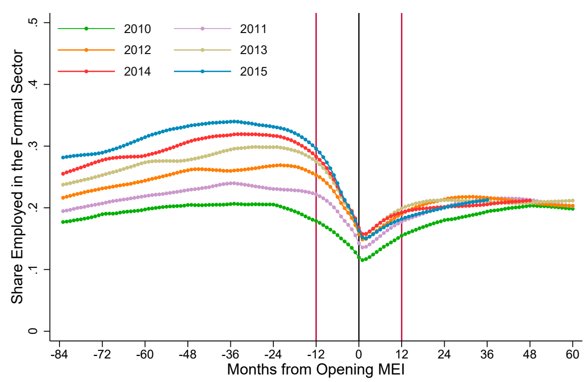
(a) Unconditional



(b) Conditional on ≥ 1 Employee

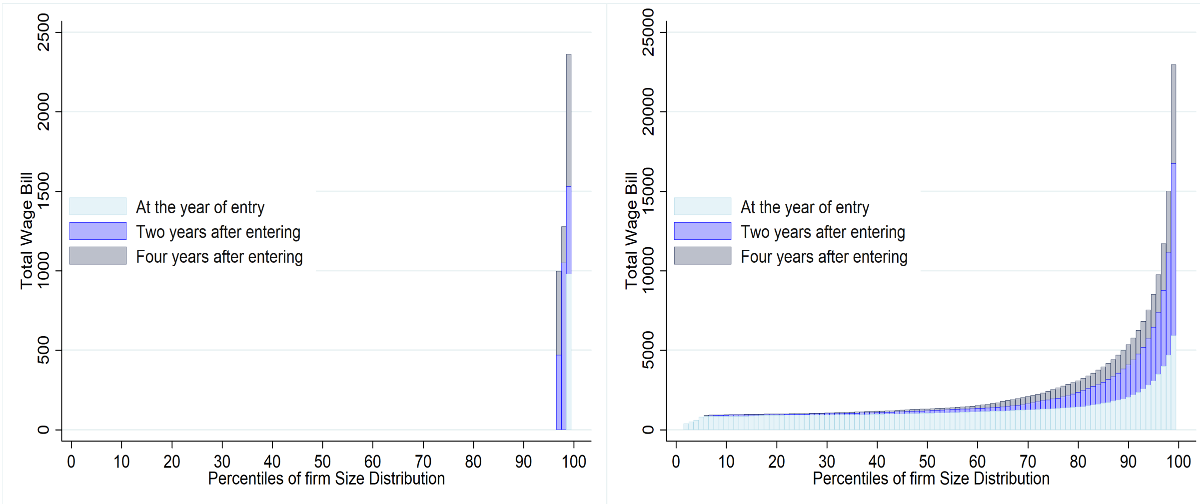
Notes: This figure is built from the panel of MEI firms combined with ownership data. Firm size is calculated as the total number of individuals that were reported in RAIS as formally employed by December 31st at the respective firm. On panel (A), we input zero workers to an active MEI firm that does not appear in RAIS. On panel (B), we restrict our sample to microentrepreneurs who appear in RAIS.

Figure A19: Share of Microentrepreneurs that were employed in the formal Sector



Notes: This figure plots the share of individuals that were employed in the formal sector relative to the month at which they opened their MEI by the year they opened their firm. The sample comprises all microentrepreneurs with a valid Social Security Number in the Tax Authority administrative records that registered between Feb. 2010 and Dec. 2015. Being employed in the formal sector means having a valid contract registered at RAIS. Vertical red lines mark one year before and after the registration as a MEI and the black line marks the month at which the individual registered as a MEI.

Figure A21: Wage Bill Distribution of Microentrepreneurs that Register as MEI



A. Wage Bill Unconditional

B. Conditional on ≥ 1 Employee

Notes: This figure is built from the panel of MEI firms combined with ownership data. The procedures to build the data set are carefully described in Appendix A.4. Wage bill is calculated as the sum of all employee's wages by December 31st at the respective firm. On panel (A), we input zero workers to an active MEI firm that does not appear in RAIS. On panel (B) we restrict our sample to microentrepreneurs that do appear in RAIS.

Additional Tables

Table A1: Share of Ineligible 7-Digit CNAE industries by Broad Industry Classification

Broad Industry Classification	Share Ineligible	Number of CNAEs
Agriculture, Livestock, Forestry, Fishing, and Aquaculture	83.93	112
Extractive Industries	97.50	40
Manufacturing	68.24	403
Electricity and Gas	0.00	8
Water, Sewerage, Waste Management, and Remediation	42.86	14
Construction	61.70	47
General Retail and Repair of Motor Vehicles and Motorcycles	65.78	225
Transport, Storage, and Mail	70.15	67
Accommodation and Food	31.25	16
Information and Communication	91.49	47
Financial, Insurance, and Related Services	100.00	60
Real Estate Activities	100.00	6
Professional, Scientific, and Technical Activities	75.00	40
Administrative and Support Services	42.59	54
Public Administration, Defense, and Social Security	100.00	9
Education	65.22	23
Human Health and Social Work Activities	98.11	53
Arts, Culture, Sports, and Recreation	81.48	27
Other Service Activities	41.67	36
Domestic Services	0.00	1
International Organizations and Other Extraterritorial Institutions	0.00	1
Total	71.92	1289

This table presents the share of ineligible CNAEs and the number of 7-digit CNAE codes by broad classification of industries. The broad classification was obtained from the official classification codes. Number of CNAES are those with at least one firm registration in 2008.

Table A2: Industries with more registrations in 2008 by MEI Eligibility

Ineligible to MEI	Eligible to MEI
Sugarcane cultivation	Manufacture of clothing (except underwear)
Construction of buildings	Street vending of miscellaneous goods
Cattle farming	Retail of pharmaceutical products
Retail of computer equipment	Retail of vehicle parts
Bars and similar establishments	Retail of bread and baked goods
Business consultancy	Retail of construction materials
Industrial maintenance services	Retail of household items
Activities of sports associations	Restaurant services
Labor unions and similar organizations	Road freight transport
Associations not elsewhere classified	Small eateries (Lanchonetes)

This table presents the ten 7-digit CNAE codes with more establishments registering in 2008 among eligible and ineligible industries.

Table A3: Summary Statistics of the MEIs in the tax authority Data

	< 2010	2010	2011	2012	2013	2014	2015	Total
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Total entry of MEI's								
	214,805	743,041	922,981	1,058,196	1,255,788	1,342,726	1,488,381	7,025,918
Panel B: Share of MEI's that have a missing Social Security Number								
	191,711	50,646	56,064	55,390	52,660	46,389	44,360	497,220
	0.892	0.068	0.061	0.052	0.042	0.035	0.030	0.071
Panel C: Share of MEI's that were active for at least 1 year								
	213,275	723,856	897,873	993,290	1,120,888	1,181,320	1,285,871	6,416,373
	0.993	0.974	0.973	0.939	0.893	0.880	0.864	0.913
Panel D: Share of MEI's by Region								
North	1,912	58,537	54,132	61,807	68,743	68,689	73,632	387,452
	0.009	0.079	0.059	0.058	0.055	0.051	0.049	0.055
Northeast	38,587	184,756	189,823	204,392	241,135	250,868	272,239	1,381,800
	0.180	0.249	0.206	0.193	0.192	0.187	0.183	0.197
Southeast	111,066	326,451	467,924	535,902	636,425	694,346	780,020	3,552,134
	0.517	0.439	0.507	0.506	0.507	0.517	0.524	0.506
South	49,981	98,657	127,764	160,027	194,951	209,978	236,686	1,078,045
	0.233	0.133	0.138	0.151	0.155	0.156	0.159	0.153
Midwest	13,259	74,639	83,338	96,068	114,534	118,845	125,804	626,487
	0.062	0.100	0.090	0.091	0.091	0.089	0.085	0.089

Notes: Table elaborated by the authors from the tax authority data set. A unique firm is defined by an unique CNPJ. Regions are divided according to the official geographic division of the country. A firm is defined as active if its official record hasn't changed to inactive or suspended.

Table A4: Estimates of Program's Effect on Firm Entry - With Matching

	(1)	(2)	(3)	(4)
	Log(Firm Entry)		Firms Entry	
Eligible x Phase 1	.43***	.43***	52***	32***
	(.034)	(.042)	(9.3)	(8.7)
Eligible x Phase 2	.96***	.92***	172***	112***
	(.052)	(.073)	(30)	(26)
Eligible x Phase 3	1.1***	1***	249***	176***
	(.067)	(.097)	(45)	(39)
Dep. Var. Mean of Treated	3.04	3.06	94.02	94.88
Month x 2 digit Ind. F.E.	No	Yes	No	Yes
Observations	80800	79278	80800	79278

This table presents estimates of coefficients $\{\beta_1, \beta_2, \beta_3, \}$ from equation 2 using a sample of matched CNAE industries. The sample consists of a monthly panel at the 7-digit industry level from 2005 to 2015 using industries matched. Match is done without replacement using a propensity score matching that estimates propensity to treatment using 2008 firm entry, number of active firms, and number of firms in the SIMPLES system. Columns (1) and (2) use as outcome the log of firm registration at a given month for a given 7-digit industry. Columns (3) and (4) show results using the number of firms registering as the outcome. Standard Errors are clustered at the industry level.

Table A5: Estimates of the Program's Effect on Firm Entry By State Groups

	(1)	(2)	(3)
	Log(Firm Entry)		
	SP,RJ,MG,DF	PR,RS,SC,CE	Remainder
Eligible x Phase 1	.52***	.28***	.013
	(.043)	(.038)	(.025)
Eligible x Phase 2	.85***	.56***	.55***
	(.063)	(.057)	(.065)
Eligible x Phase 3	1.1***	.77***	.63***
	(.083)	(.073)	(.077)
Dep. Var. Mean of Treated	1.66	1.27	.84
State x Month x 2 digit Ind. F.E.	Yes	Yes	Yes
Observations	308388	322559	619446

This table presents estimates of coefficients $\{\beta_1, \beta_2, \beta_3\}$ from equation 2. Observations are at the State × Month × Industry level. The sample is divided across columns by states according to the month in which they were allowed into the program. The column headers show the state abbreviations for each group. Standard errors are clustered at the 7-digit CNAE industry level.

Table A6: Summary Statistics of MEI work Activity

	2011	2012	2013	2015
	(1)	(2)	(3)	(4)
Location of Work:				
Works at: Street	0.180	0.120	0.104	0.130
Works at: Office	0.390	0.340	0.302	0.280
Works at: Home	0.400	0.430	0.486	0.530
Works at: other's Firms	.	0.110	0.107	0.0700
Works at: Other (2011)	0.0300	.	.	.
Sector of Work:				
Sector: Construction	0.0730	0.0800	0.0880	0.0950
Sector: Retail	0.395	0.390	0.393	0.374
Sector: Manufacturing	0.176	0.170	0.147	0.153
Sector: Services	0.356	0.360	0.367	0.372
Motive for Registering as MEI				
Motive: Formal Firm Benefits	0.630	0.690	0.785	0.680
Motive: Social Security Benefits	0.370	0.310	0.215	0.320
Had Other Income Sources	0.220	0.260	0.240	0.230
Where you active as MEI at the moment of the survey				
Yes	-	0.90	0.83	0.88
No	-	0.10	0.17	-
No - Ended Business	-	-	-	0.08
No - Still haven't Started	-	-	-	0.03
No - Outgrown MEI	-	-	-	0.01
Observations	10585	11577	12000	9657

Notes: This table is elaborated from information extracted from Perfil do MEI surveys. Their sample is a random sample from individual microentrepreneurs registered at the federal tax authority.

Table A7: Effects on Social Security Benefits - Type of Benefit

	(1) All	(2) Health	(3) Maternity	(4) Pensions	(5) Retirement
Eligible x Phase 1	.024	-.086	.35*	-.0032	.038
	(.11)	(.084)	(.2)	(.096)	(.12)
Eligible x Phase 2	.096	.024	.51***	-.008	.02
	(.072)	(.051)	(.14)	(.067)	(.077)
Eligible x Phase 3	.31***	.11***	.96***	.04	.12**
	(.044)	(.034)	(.085)	(.04)	(.048)
Dep. Var. Mean of Treated	7.35	6.63	4.88	5.17	6.11
Observations	2376	2376	2370	2376	2376

Notes: This table presents estimates of the effects of the MEI program on the types of benefits issued. Column (1) replicates the main estimates using all benefits. Columns (2) use as outcome benefits given due to health reasons. Column (3) shows results using as outcome maternity benefits. Column (4) show results using pensions and Column (5) retirement benefits. The benefit-level data is aggregated at the State x Month x Class.

A Additional Data Description

In this appendix, we describe the details of the CNPJ data used in our analysis and thoroughly discuss the differences between CNPJ and RAIS data. We then describe additional data sets used, and construction of different analysis data.

A.1 The CNPJ data

The *Cadastro Nacional da Pessoa Jurídica* (CNPJ) is the administrative database under the responsibility of *Receita Federal do Brasil* (RFB) that holds registration information of entities (e.g., business partnerships, individual entrepreneurs, public agencies, among others) of interest to the tax administrations of the federal and local governments in Brazil. It is updated constantly by the *Coordenação Geral de Gestão de Cadastros* (Cocad), the coordination responsible for taking the administrative procedures to publicly notify entities with register irregularities. And it is maintained by the *Serviço Federal de Processamento de Dados* (SERPRO), a data processing public company that provides technology services to the Brazilian federal government.

The CNPJ database has two parts: registration information of the entities in CNPJ; and, registration information of partners and administrators of some types of entities in the database. It used to be a confidential dataset to which access had to be granted by RFB which hardly ever authorized it. Eventually, the matter became judicial and the *Controladoria Geral da União* (CGU) ended up deciding - in late 2018 - in favor of the claimants, determining that the database should be open to all Brazillians[37]. Ever since then, the RFB publishes quarterly, in its website,

[37] See http://dados.gov.br/noticia/cgu-libera-acesso-a-base-de-dados-de-cnpj-da-rfb .

the last version of the data processed by SERPRO withholding the CPF numbers of the partners and administrators in the dataset.

Data Quality - Firm Activity

Asides regulatory and fiscal obligations, entities are obliged to maintain registration data up to date in Brazil[38]. Cocad has the authority to restrain entities' registration statuses from being active in case of repeated failure to comply with fiscal obligations, inconsistencies in the registration information, amongst several other occasions[39]. For instance, if an entity fails to present any mandatory statement for two consecutive years[40] or fails to confirm receiving two or more tracked mail from RFB, the Cocad can publicly summon the entity by notice and set its registry status as inapt. An important example rather not related to compliance failure is when an entity is in the process to close its register with RFB, its registry status is set to suspended throughout the whole operation.

On such occasions, Cocad not only updates the entities' registry statuses but also records the date of the change in CNPJ as well. This fact, along with the entities' date of start of the activity, provided us the beginning and the end of activity spell of each entity in CNPJ which allowed us to filter them in regard to such information. This way, we only keep in our sample the entities that were active between January 2005 and December 2015 in CNPJ.

A register can be closed either by the entity's responsible requests or administratively, by the RFB. There are five situations for a company to get administratively closed by the RFB: notified by repeated omission; notified by administratively inexistance; inapt not regularized in 5 years; with cancelled registration by the register agency; or, by court order. The former three occur when detected lack of comply of obligations. Default omission is the entity that, being obliged to, has not submitted, for 5 (five) or more years, none of the statements.

The CNPJ register is suspended[41] mainly when: the entity's responsible place a closing request to RFB and it is either under analysis or if it has been rejected; it is declared legally inexistent[42]; temporarily interrupted its activities and have declared this situation to RFB; it is declared legally an insistently non-complier, which is one that, being obliged to, has not submitted, for 5 (five) or more years, none of the statements[43]; or, have inconsistencies in the registration data[44]. In the last two cases, the Cocad publicly notifies the entities, that has to regularize their situation with RFB within a specific time, otherwise the register is closed. Thus there is an active administrative control over the registries in the database, giving some reliability to the data.

[38] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 24.
[39] See *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 38-51.
[40] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 29, item I.
[41] See Art 40, Art 29 I and II for a full description of the situations that can get the register suspended.
[42] Art 29 Inciso II
[43] Art 29 Inciso I
[44] Art 40 Par 2o.

The CNPJ register gets inapt when: omission of declarations and statements, thus considered the one that, being obliged, fails to present, in 2 (two) consecutive years, any of the declarations and statements[45] ; failed to receive 2 tracked correspondences from RFB; with irregularities in foreign trade operations, thus considered one that does not prove the origin, availability and effective transfer, if applicable, of the resources used in foreign trade operations, as provided by law. In the case of an inapt register, the entity suffers sanctions such as it is included in the Informative Register of Unpaid Credits of the Federal Public Sector (Cadin); it is prevented from participate in public purchase auctions, enter into covenants, agreements, adjustments or contracts that involve disbursement, in any capacity, of public resources, and respective amendments; obtain tax and financial incentives; carry out credit operations that involve the use of public resources; and, transact with banking establishments, including with regard to the movement of current accounts, the realization of financial investments and the obtaining of loans. Hence, there are incentives for the responsible to keep the registration information up to date so they do not get their registered inapt.

Other Registration statuses are active, suspended, inapt and close. We use the date of opening and the date of change in registration status to other than active to determine the establishments that were active in a determined time frame. We keep only the establishments active between 2005 and 2015.

Data Accuracy

The law provision determining the responsibility on the registration data accuracy to the entity is not the only existent instrument to enforce compliance towards registration obligations. Outdated registration information and non-active registry status can have harsh consequences on the entity according to the regulation. Thus, giving them actual incentives to comply with their registration responsibilities.

Inconsistent partners and administrators' data can prevent entities from closing their registries and end obligations along with the tax authority[46]. Additionally, an entity with an inapt registry loses the validity of its invoice on any tax effects in favor of an interested third party[47] and gets prevented from: transacting with banking establishments, including the handling of checking accounts, making financial investments and getting loans; obtaining fiscal and financial incentives; carrying out credit operations that involve the use of public resources; participating in public auctions to provide services for the government; and, entering into covenants, agree-

[45] listed in item I of the caput of art. 29

[46] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 28.

[47] According to *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 48. the invoice from an inapt registry company loses validity on being: deducted as cost or expense, in determining the calculation basis for Corporate Income Tax (IRPJ) and Social Contribution on Net Income (CSLL); deducted in determining the basis for calculating the Personal Income Tax (IRPF); used as credit for the Non-cumulative Tax on Industrialized Products (IPI), the Contribution to PIS/Pasep and the Contribution to the Financing of Social Security (Cofins); used to justify any other deduction, reduction, reduction, compensation or exclusion related to the taxes administered by the RFB.

ments, adjustments or contracts that involve public resources[48]. All of which corroborates in favor of the CNPJ being a reliable administrative data source to study formal entrepreneurship in Brazil.

The procedure to enroll or modify registration information in CNPJ also goes in the same direction. In order to perform any registration act[49], one must go through a procedure validated by RFB. First, the registration act request must be submitted online through a website administered by RFB[50]. Then, RFB makes available several documents related to registration information in the CNPJ for filling[51]. Once the documentation is filled out it must be submitted to RFB through that same website. RFB, then, screens the filled documentation against any data inconsistencies. It is only after going successfully through RFB's screening that the form for the registration act is made available to the requester. With that in hand, the form must be filled out and signed by the person responsible for the entity. The registration act is formalized once that form is delivered to RFB along with a list of additionally required documentation.

The responsible can change registration data in terms of the head office's corporate name; legal nature; social capital; the size of the company; the entity's representative in the CNPJ; to the representative; the partners and administrators data; to the responsible federative entity, in the case of Public Administration entities; bankruptcy; judicial recovery; intervention; the inventory of the individual entrepreneur or the holder of an individual real estate or limited liability company; judicial or extrajudicial liquidation; incorporation; fusion; partial or total spin-off[52]. Importantly, although changes in legal nature and industry are possible, if made by an establishment participating in MEI out of the individual entrepreneur legal nature, into an industry not allowed in the program or opening a branch establishment, that establishment is automatically excluded from the program[53].

A.2 MEI firms reporting to RAIS or CNPJ?

RAIS is the well-established matched employer–employee administrative dataset collected annually by the Ministry of Labor. It is considered a high-quality source because establishments have strong incentives to accurately report, given that RAIS underlies the official calculation of social security contributions and multiple labor audits. Importantly, prior to the creation of the MEI regime, most registered establishments—regardless of whether they had employees—were required to submit RAIS information annually. Those that do not have employees appear only

[48] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 46

[49] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 13 states that the following are registration acts in the CNPJ: registration; alteration of registration data and registration status; closing; reinstatement of registration; and declaration of nullity of the registration act.

[50] http://www.redesim.gov.br/

[51] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art 14, the documents shared when a registration act request is first submitted are : Legal Entity Registration Form (FCPJ); the partners and administrators registration form; Specific Form of the agreement; Final beneficiaries file.

[52] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 17.

[53] *Resolução CGSN no 94/2011 de 29 de novembro de 2011*, Art. 105.

in an establishment level data, often referred to as RAIS negativa, which is the one used in Rocha et al. (2018).

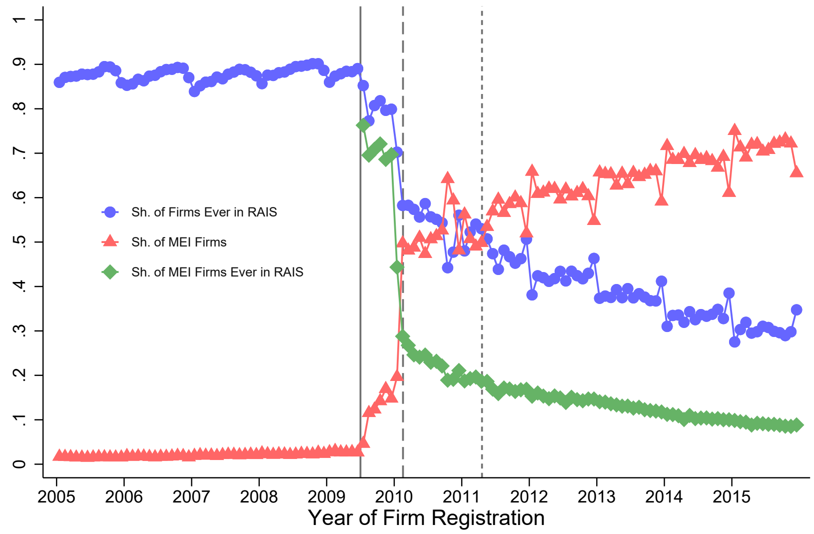
However, MEI firms that do not employ were exempted from reporting to RAIS. Initially, it was unclear if the MEI with no employee would have to report the RAIS Negativa. In 2010, some MEI firms were notified by the authorities charging a fine for not reporting the RAIS, but in April 2010 the Labor Ministry made an official statement exempting them from the charges (github same link). In February 2011, the Labor Ministry officially determined the exemption (github stable link). In November 2011, a change to the MEI Law clearly stated the exemption from reporting the RAIS.

In turn, all firms—including MEI—must register in the CNPJ. The CNPJ is the federal tax authority's national registry of legal entities, and it covers the universe of formal firms in Brazil regardless of tax regime, size, or whether the firm ever hires an employee. MEI status is granted only after the entrepreneur obtains a CNPJ identifier, and maintaining an active MEI requires remaining in good standing in the CNPJ system. Therefore, unlike RAIS—whose reporting obligation does not apply to MEIs without employees—the CNPJ registry systematically captures the entire population of MEI firms.

These differences are presented in Figure A22. We take the universe of CNPJ-registered firms that entered between 2005 and 2015 (excluding rural producers and political organizations) and match them to all establishments that report in RAIS Negativa from the calendar year of their creation, to 3 calendar years after. The blue circles plot, for each month of registration, the share of all newly registered firms that ever appear in RAIS Negativa. The green line plots the analogous share for MEI firms only. To help illustrate our point, we also plot, in red triangles, the share of MEI among all newly registered CNPJ firms.

This figure serves three purposes. First, it validates the use of CNPJ as a universe frame: prior to the introduction of MEI, the vast majority of newly registered firms eventually appear in RAIS, which confirms that CNPJ-based measurement is consistent with the pre-MEI institutional environment. Second, the sharp increase in the share of CNPJ firms that never show up in RAIS aligns exactly with the timing of the MEI program—evidence that the rise of "non-RAIS firms" is not measurement noise, but rather a mechanical consequence of the new legal exemption. Third, the figure highlights that the majority of MEI firms themselves never appear in RAIS. This is already visible in February 2010—the first cohorts of MEI entrants—and remains extremely low throughout. This reinforces our central argument: RAIS misses precisely the margin where the MEI operates, while CNPJ preserves coverage of that population.

Figure A22: Share of Firms that Report to RAIS



This Figure summarizes statistics of firms in the CNPJ data set reporting to RAIS. Sample comprises all CNPJ data firms that entered registered between 2005 and 2015, excluding rural producers and political organizations. We match them to all firms reported to RAIS Negativa (establishment level data that includes non-employers) between the calendar year of their creation, and 3 calendar years after. The blue circles represent the share of all firms that registered in that given month appearing in RAIS negativa. The red triangles show the share of MEI firms among all CNPJ firms registering in a given month. The green diamond markers show the share of MEI firms that registered in that given month who ever report to RAIS Negativa between 2003-2020.

A.3 Additional Data Sets

In the paper, we use other data sources not directly described in the main body of the text. Here we describe them in detail:

Relação Anual de Informações Sociais (RAIS) is an administrative registry maintained by Brazil's Ministry of Labor. RAIS covers the universe of formal employer–employee relationships in the country and records detailed information on every worker employed under a formal contract at any point during the calendar year. For each job spell, the dataset includes individual characteristics (e.g., age, gender, education), monthly earnings, occupation codes, hours worked, establishment identifiers, and firm characteristics such as industry, location, and legal form. Because reporting is compulsory for all establishments with at least one formally registered worker, RAIS provides complete coverage of the formal labor market and offers a consistent longitudinal structure, allowing workers and firms to be tracked over time.

Cadastro Nacional de Empresas (CNE): The CNE is a dataset formerly organized by the Ministry of Industry and Commerce (MDIC) that compiles firm registrations from state-level business registries. It was collected until 2017, and access is restricted; we obtained the data through direct contact with MDIC. Every business establishment in Brazil must register with both federal and state authorities. The state-level entities—known as Juntas Comerciais—collect information similar to that reported to the Federal Tax Authority. The CNE complements the CNPJ data by providing the business name and the full CPF of single-owner firms.

Perfil do MEI – SEBRAE: These data come from telephone surveys conducted by SEBRAE, an autonomous public agency responsible for supporting small firms and microentrepreneurs in Brazil. Between 2011 and 2015, SEBRAE published annual reports based on interviews with a randomized, representative sample of MEI firm owners, collecting detailed information on their activities. Although the underlying microdata are not available to us, we scrape and compile the published reports to extract descriptive statistics—such as reasons for formalization and access to credit—among MEI-registered firms.

The INSS data

The *Instituto Nacional do Seguro Social* (INSS) was established on June 27, 1990, through the merger of the Institute of Financial Administration of Social Security and the Assistance and the National Institute of Social Security. It is an autonomous agency that operates under the Ministry of Social Security and is responsible for recognizing entitlements and ensuring citizens' access to social security benefits, such as retirement, pensions, and maternity leave. We use two datasets from the INSS that are publicly available: the Beneficios Mantidos dataset and the *Anuário Estatístico da Previdência Social* (AEPS). In the following, we describe each dataset.

Beneficios Mantidos: Data on all benefits ever paid by the INSS. It includes active, terminated, and suspended benefits, in accordance with Decree No. 8,777/2016 and the Access to Information Act No. 12,527/2011. The data is extracted from the Unified Benefit Information System (SUIBE) and includes information on benefit types, medical condition codes (CID), beneficiary classification, gender, affiliation type, reason for termination or suspension, status group, municipality, state (UF), dependent relationships, beneficiary's date of birth, benefit termination date (DCB), benefit approval date (DDB), benefit start date (DIB), economic sector, benefit category, and monthly income. We follow the INSS classification to group beneficiaries by their affiliation type. The INSS categorizes affiliates into two main groups: "Employed Contributors" and "Other Contributors." Within the "Other Contributors" category, there is the Individual Contributor classification, which comprises entrepreneurs, self-employed workers, and those classified as equivalent to self-employed. Microentrepreneurs registered under the MEI program fall within this category.

AEPS: Social Security yearly reports on the system's figures in revenues and expenditures, the number of contributors, benefit flows and stock, and social security coverage. This data allows us to directly observe the revenues and the number of contributors from the MEI program, which we use to discuss the benefits of formalization to the government revenue. The INSS numbers reported are organized in two groups of contributors: Employed and Other Contributors. Employed contributors are the wage-dependent workers, except for domestic workers. Other contributors include the Individual Contributor, Domestic Worker, Facultative Contributor, and Special Insured. The Individual Contributor provides services of an urban or rural nature, on an occasional basis, to one or more companies, without an employment relationship; or engages in a remunerated economic activity of an urban or rural nature, whether for profit or not. It is the affiliation of entrepreneurs and self-employed. The Domestic Employee provides continuous service, with monthly remuneration, to a person or family, in a non-profit activity. The Special Insured is an individual who resides in a rural property or in a nearby urban or rural settlement and who, either individually or under a family-based economy, engages in agricultural activities on land of up to four fiscal modules; or who works as a rubber tapper or plant extractor, making these activities their primary means of livelihood; or as an artisanal fisherman, making fishing their regular profession or main source of income; or the spouse, partner, or child over 16 years old of the insured person who works with the family group. The Facultative Contributor is an individual over 16 years of age who joins the General Social Security System by making contributions.

A.4 Creating a Panel of Firm Level Growth for Each Entrepreneur

Combining our three administrative data sets (CNPJ, CNE and RAIS) we create a panel that tracks firm size (measured as number of formal employees) of all microentreprises registered in the MEI tax system between 2009 and 2015 as well as other firms owned by microentrepreneurs that registered MEI firms. Here we detail how we created this data set.

The RAIS data set contains the universe of the formal labor market. From it, we can construct the number of formal employees at all formal establishments in Brazil. Establishments in RAIS are identified by the same unique identifier as in the CNPJ and CNE data sets. As we described in sections 2 and A, MEI firm owners that do not have an employee don't need to report RAIS information to the ministry of labor, but the ones that do employ one worker must report it. We thus input zero employees to MEI firms that are active in CNPJ data but are not present in RAIS.

We can thus observe for each MEI firm, the number of workers and wage bill in every year that the firm is active. This would however be only a first order approximation of firm growth because anecdotal evidence suggests that some entrepreneurs prefer to register a new firm ID when they grow out of the MEI program. If this anecdotal evidence represents a large share of the MEI firms that grow, only matching MEI firms to RAIS through their unique firm identifier would underestimate the growth of MEI firms.

To overcome this issue, we use our two other administrative data sets, CNE and CNPJ to recover ownership information of firms[54]. Table A8 summarizes our firm ownership data. Each column corresponds to a year of firms in RAIS. The first row shows the share of firms that we have information on who the owners are. For most years of the sample we have more than 90% of firm owners. We also see that the firms that we do not have information are on average much smaller than the ones that we do know who the owners are.

With our ownership information we organize our data set at the microentreprenuer level, instead of the microentreprise level. Thus we take into account both the MEI firm that she owns and other possible firms that she opens. If she opens a regular firm after her MEI firm, the new firm is taken into account as the firm size for that microentrepreneur. If she has more than one firm than we sum the sizes of these two firms. If the owned firm has more than one partner, we input the same number of employees to all partners, not adjusting by number of partners[55]. Similar procedures are made with the wage bill, which we define as the total amount of wages paid in December of the respective year.

[54]To our understanding, that was the same data procedure used in Colonnelli et al. (2020).

[55]One could argue that such adjustment is needed. For example, in a firm with 5 partners, one could think that we should divide the number of employees by five and input this share as the number of workers to the respective microentrepreneur. As our results point to the lack of growth, we believe we are conservative in our measure, and we are actually overestimating the number of employees under the supervision of thes microentrepreneurs.

Table A8: Summary Statistics from Ownership Data 2009-2018

	(1) 2009	(2) 2010	(3) 2011	(4) 2012	(5) 2013	(6) 2014	(7) 2015	(8) 2016	(9) 2017	(10) 2018
Sh. with Ownership Info	0.936	0.931	0.926	0.923	0.918	0.916	0.915	0.914	0.907	0.888
Sh. with CNE Own Info	0.784	0.777	0.769	0.765	0.755	0.751	0.742	0.734	0.714	0.670
Sh. with CNPJ Own Inco	0.702	0.696	0.691	0.685	0.678	0.671	0.672	0.674	0.674	0.672
Sh. With Own Info excluding RJ and MG	0.943	0.939	0.935	0.933	0.929	0.928	0.930	0.931	0.924	0.904
Sh. CNE not RJ or MG	0.850	0.845	0.840	0.836	0.828	0.826	0.820	0.815	0.796	0.747
Sh. Receita not RJ or MG	0.683	0.677	0.672	0.667	0.661	0.655	0.656	0.658	0.659	0.658
Avg. Wage Bill (all firms)	33489.5	34645.9	35836.9	36586.7	37321.1	36834.4	34182.1	33248.6	33793.7	34166.0
Avg. Number of Workers (all firms)	14.08	14.34	14.38	14.21	14.03	13.73	13.16	12.74	12.74	12.97
Avg. Wage bill (firms with Own. Info)	35394.9	36812.1	38256.5	39156.2	40156.8	39709.4	36862.6	35870.6	36707.3	37811.8
Avg. Wage bill (firms w/o Own Info)	5424.6	5446.4	5469.1	5622.3	5655.7	5624.2	5356.2	5317.9	5415.0	5281.6
Avg. Number of Workers (firms with Own. Info)	14.79	15.14	15.25	15.10	14.97	14.68	14.07	13.63	13.71	14.20
Avg. Number of Workers (firms w/o Own. Info)	3.633	3.581	3.558	3.548	3.452	3.411	3.356	3.263	3.254	3.230
Observations	1994586	2139009	2263606	2375698	2478617	2564489	2571444	2534032	2520734	2511241

Notes: This table shows summary statistics from the firm ownership data that we created. The data set was constructed by combining CNPJ, CNE and RAIS data described in section 2. The denominator is the total number of business establishments that hire at least one worker in RAIS at year t. A business establishment is defined by having *Natureza Juridica* between 2000 and 2999. Each column restricts the sample of business establishments for the ones that are active at year t where t is the year described in the top of the table. Wage bill is the average wage times the average number of workers.

B Additional Results using PME

In this Appendix, we provide additional results using the *Pesquisa Mensal do Emprego* (PME), the household survey data employed in Rocha et al. (2018).

PME (Pesquisa Mensal de Emprego): PME is a monthly household labor force survey conducted by IBGE that covered the six largest metropolitan regions of Brazil (São Paulo, Rio de Janeiro, Belo Horizonte, Porto Alegre, Recife, and Salvador) until it was discontinued in 2015. The survey sampled roughly 38–40 thousand households and about 100–110 thousand individuals per month. Its rotating panel structure allows researchers to follow households for several consecutive months, making it particularly useful for analyzing short-run labor market dynamics and adjustment margins in urban settings.

There is a clear trade-off between using PME and PNAD. PME's rotating panel structure allows for individual-level analysis following the same households over time. On the other hand, industry classification in PME is available only at the 2-digit level, whereas program eligibility is defined at a more disaggregated level. As a result, it is not possible to construct a direct eligibility indicator within the PME dataset. Rocha et al. (2018) therefore rely on PNAD to construct industry eligibility, and then use a crosswalk between PNAD and PME industry classifications to assign eligibility in the PME data.

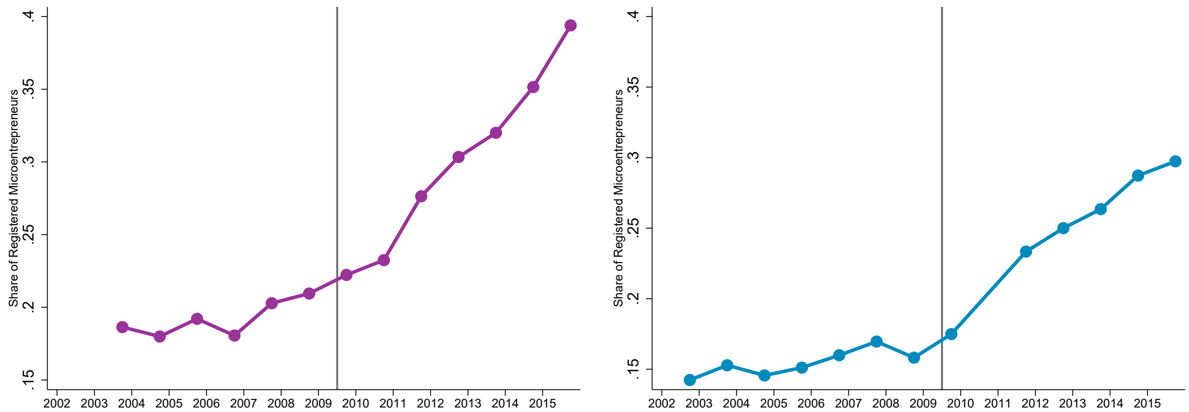
In this Appendix, our first goal is to examine whether the aggregate patterns we document using PNAD are also present in PME, and to discuss the individual-level regressions in Rocha et al. (2018).

We begin by replicating the share of microentrepreneurs who are formal using PME data. Our definition of formality is contributing to the social security system through one's job[56]. We define a microentrepreneur as an individual who declares being self-employed. In PME, we cannot restrict the sample to employers with at most one employee, as we do in PNAD, and therefore replicate our PNAD findings using this slightly broader definition of microentrepreneur.

The share of formal microentrepreneurs in PME is extremely similar to the corresponding values in PNAD. Figure A23 shows the share of formal microentrepreneurs between 2003 and 2015 in PME and PNAD, using a consistent definition of microentrepreneur across both surveys. Levels are somewhat higher in PME, which is expected given its coverage of large urban areas, where informality is, on average, lower.

[56] In PME, this corresponds to variable V425 == 1

Figure A23: Share of Formal Microentrepreneurs PME and PNPAD



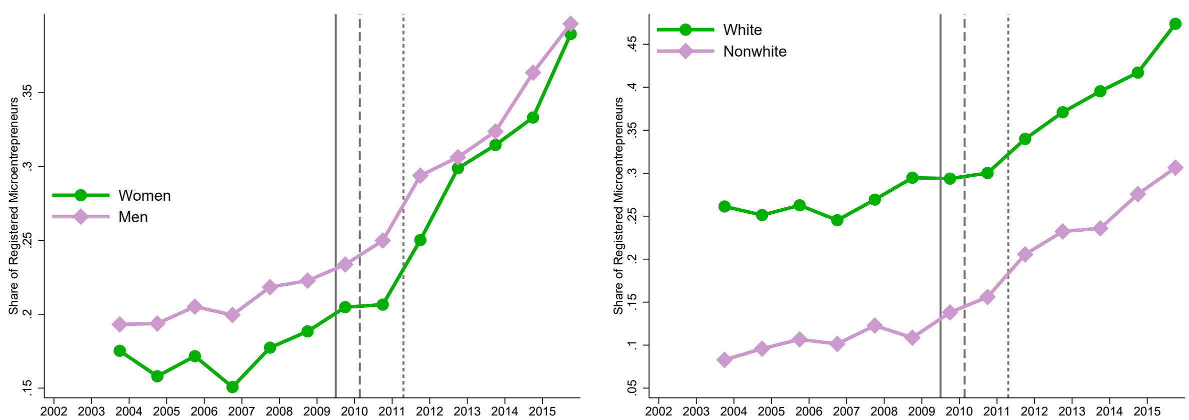
A. PME

B. PNAD

Notes: Panel A uses PME data and B uses PNAD data. Sample is restricted to microentrepreneurs, which are defined as individuals that declare themselves self-employed or firm owners with at most one employee. Sample is restricted to individuals between 18 and 60 years. Being formal is defined as contributing to Social Security Network.

As a robustness check, we also examine heterogeneity by gender and race, variables that are directly comparable across both surveys. We present these results in Figure A24. We observe patterns very similar to those shown in Figure A13, which is based on PNAD data.

Figure A24: Share of Formal Microentrepreneurs by Group including Agriculture - PME survey



A. Gender

B. Race

Notes: Both panels are built from PME data set from september of each year. Sample is restricted to microentrepreneurs, which are defined as individuals that declare themselves self-employed. Sample is restricted to individuals between 18 and 60 years. Being formal is defined as contributing to Social Security Network. Panel A divides microentrepreneurs iby gender. Panel D plots the share of microentrepreneurs years that are formal by race. Vertical lines represent the start of the different phases of the program.

Individual-level regressions in Rocha et al. (2018): One advantage of using PME is the possibility of following individuals over time. Rocha et al. (2018) exploit this feature to study dynamic responses to the MEI program and argue that its effects emerge primarily after May 2011.

In what follows, we pursue two objectives. First, we clarify that the identifying variation used in Rocha et al. (2018) arises mainly from the period between July 2009 and February 2010, which corresponds to what we define as the first phase of the program, when registration costs remained relatively high and the main simplification mechanisms had not yet been fully implemented. Second, we show that the resulting estimates are sensitive to alternative specification choices, with relatively small changes in the estimating equations leading to noticeable differences in the magnitude of the estimated effects.

What is the variation used in Rocha et al. (2018)?

First, they define an *intensity of treatment* variable using PNAD–PME industry crosswalks. It is constructed as follows in their paper:

$$Intense_s = \frac{\sum_{k \in s} \left(\mathbb{I}[\text{Sub–Industry } k \text{ is eligible}] \times N_{ks} \right)}{\sum_{k \in s} N_{ks}} \tag{7}$$

where N_{ks} denotes the number of entrepreneurs that belong to sub–industry k within the 2-digit industry s in 2009. The variable $Intense_s$, calculated with PNAD data, is then merged at the industry s level with our monthly panel of individuals from PME.

Rocha et al. (2018) use the term IMP (Individual Microentrepreneur Program) to refer to MEI. Thus, in this appendix, we will follow their terminology. Their individual level specification is:

$$\begin{aligned} Y_{irt} = {} & \beta_1 (Post1_{rt} \times Intense_s) + \beta_2 (Post2_{rt} \times Intense_s) \\ & + \gamma_1 Post1_{rt} + (Post1_t \times Intense_s)' \gamma_2 \\ & + \lambda_t + \phi_i \\ & + Trend^s_{rt} \theta + X'_{irt} \psi + \varepsilon_{irt} \end{aligned} \tag{8}$$

In their specification, the post-policy indicators are defined at the metropolitan-region-by-month level. Specifically, $Post1_{rt}$ turns on when access to the first phase of the program (IMP1) becomes available in region r at month t. Because IMP1 started on different dates across regions, $Post1_{rt}$ is implemented as a set of start-date dummies in the PME monthly data: $\mathbf{1}[t \geq \text{Aug/2009}]$, $\mathbf{1}[t \geq \text{Sep/2009}]$, and $\mathbf{1}[t \geq \text{Feb/2010}]$. The variable $Post2_{rt}$ is defined analogously for the second phase (IMP2), turning on from its start date (April 2011) onward in region r.

For visualization purposes, we can actually rewrite the equation above, disentanling the vector $(Post1_t \times Intense_s)$:

$$
\begin{aligned}
Y_{irt} = \beta_1(Post1_{rt} \times Intense_s) &+ \beta_2(Post2_{rt} \times Intense_s) \\
&+ \gamma_1 Post1_{rt} \\
&+ \gamma_{2a}(Post_{t \geq Aug/2009} \times Intense_s) \\
&+ \gamma_{2b}(Post_{t \geq Sep/2009} \times Intense_s) \\
&+ \gamma_{2c}(Post_{t \geq Feb/2010} \times Intense_s) \\
&+ \lambda_t + \phi_i \\
&+ Trend^s_{rt}\theta + X'_{irt}\psi + \varepsilon_{irt}
\end{aligned} \tag{9}
$$

Rewriting the equation makes it clear that β_1 is identified only from the staggered rollout of IMP1 across metropolitan regions. Because different metros gained access to the program in July 2009, September 2009, or February 2010, identification comes from comparing changes in formalization between regions that had already adopted IMP1 and those that had not yet done so, within the same month, weighted by industry-level exposure intensity. After February 2010, when all regions became eligible, this cross-regional timing variation disappears. From that point onward, any remaining differences across high- and low-intensity industries are captured by the calendar-time $\times$ intensity interactions (e.g., γ_{2c}), which absorb common post-adoption shifts rather than contributing to the identification of β_1.

Thus, β_1 is identified by exploring two different comparisons. First, variation of treated and control industries in SP,MG,RJ versus RS, BA and PE between July 2009 and August 2009, and variation of treated and control industries in SP,MG,RJ and RS versus BA and PE between Sept. 2009 and February 2010. All variation between treated and control industries after February 2010 is absorbed by $\gamma_{2c}(Post_{t \geq Feb/2010} \times Intense_s)$.

As we show in our manuscript, in this staggered implementation (between July 2009 and February 2010), there are limited effects of the program, particularly because the entry costs are still substantially high, and the simplified website had not been released yet.

How Sensitive are the Estimates from Rocha et al. (2018)?

They estimate equation 8 using a panel of microentrepreneurs from all metropolitan regions covered by the PME survey. We use their publicly available replication files to reproduce their main specifications and estimate closely comparable regressions. Although not explicitly described in the main text, the replication materials also exclude a set of month–metropolitan-region cells that coincide with the initial implementation months of IMP1 in each metro, as well as the first two months of IMP2. These omitted observations correspond to what can be interpreted as transition periods between the pre-policy and post-policy regimes for each staggered start date. Specifically, for Belo Horizonte (31), Rio de Janeiro (33), and São Paulo (35), observations in July and August 2009 are excluded; for Porto Alegre (43), September and October 2009 are excluded; and for Recife (26) and Salvador (29), January and February 2010 are excluded. In addition, the months of April and May 2011 are excluded for all metropolitan regions, corresponding to the start of the second phase (IMP2).

In our first set of results, we replicate the main analysis of Rocha et al. (2018) and then estimate the same specifications without excluding these transition periods. Columns (1)–(3) of Table A9 reproduce their baseline results, which correspond to Table 4 in their paper. They interpret the near-zero estimate for the first phase as evidence of limited effects of the MEI program on formalization prior to May 2011. We find that simply retaining the transition periods leads to substantially larger point estimates for the first phase, indicating that the magnitude of the estimated effects is sensitive to relatively small specification choices.

Table A9: Rocha et al. (2018) including dropped periods

	(1)	(2)	(3)	(4)	(5)	(6)
	Being a Formal Entrepreneur					
	Original Results			Without Dropping Transition Periods		
IMP1	0.00325	0.00527	-0.000705	0.0181	0.0213	0.0199
	(0.0336)	(0.0318)	(0.0258)	(0.0214)	(0.0215)	(0.0220)
IMP2	0.0669*	0.0684*	0.0440***	0.0562*	0.0576*	0.0363***
	(0.0317)	(0.0322)	(0.0117)	(0.0262)	(0.0268)	(0.00922)
Basic DD Spec.	Yes	Yes	Yes	Yes	Yes	Yes
Control Econ. Fluc.	No	Yes	Yes	No	Yes	Yes
Control Convergence	No	No	Yes	No	No	Yes
N	27907	27907	27907	31753	31753	31753

Notes: Columns (1)-(3) are replication of Rocha et al. (2018). They estimate three specifications with progressively richer sets of controls. Specification (1) includes time and entrepreneur fixed effects, as well as individual covariates (age, age squared, and schooling) and interactions with treatment timing. Specification (2) adds controls for local economic conditions and demographics—such as regional wages, hours worked, employment, gender composition, education, and age—along with higher-order terms. Specification (3) further includes convergence controls by interacting the baseline outcome (measured in 2006 at the region–industry level) with polynomial time trends. Columns (4)-(6) are the same exact specifications without dropping transition periods.

The results discussed above do not directly address the role of the identifying variation described earlier. We therefore turn to that issue next.

In Table A10, we estimate different versions of their first specification, but excluding the $Post1_t \times Intense_s$ controls. Column (1) replicates their first result, column (2) drops from the regression the variable ($Post_{t \geq Feb/2010} \times Intense_s$), and column (3) drops from the estimation the whole vector $Post1_t \times Intense_s$. In columns (4) and (5) we do the same, but including the originally excluded transition periods.

We observe that the estimates of the effect of the program in its first phase are sensitive to the inclusion or exclusion of $Post1_t \times Intense_s$. In contrast, the estimates for the second phase (IMP2) remain largely unchanged across specifications, as expected.

Table A10: Rocha et al. (2018) without $Post1_t \times Intense_s$

	(1)	(2)	(3)	(4)	(5)
	Being a Formal Entrepreneur				
IMP1	0.00325	0.0130	0.0346	0.0240	0.0343
	(0.0336)	(0.0226)	(0.0220)	(0.0161)	(0.0202)
IMP2	0.0669*	0.0669*	0.0669*	0.0563*	0.0563*
	(0.0317)	(0.0317)	(0.0317)	(0.0262)	(0.0262)
N	27907	27907	27907	31753	31753

Notes: Column (1) replicate Rocha et al. (2018) first result, column (2) drops from the regression the variable ($Post_{t\geq Feb/2010} \times Intense_s$), and column (3) drops from the estimation the whoe vector $Post1_t \times Intense_s$. In columns (4) and (5) have the same results, but including the originally excluded transition periods.

The sensitivity to the inclusion of $Post1_t \times Intense_s$ reflects the staggered implementation of the program across regions. To get the cleanest estimates that do not deal with this, we can restrict ourselves for the first set of states that got the MEI program, and estimate a more *standard* difference in differences design. We do so in Table A11, where we limit the sample to São Paulo (SP), Rio de Janeiro (RJ), and Minas Gerais (MG)—the regions in the PME data that entered the first phase of the MEI. In this restricted sample, we estimate equation 8 without $Post1_t \times Intense_s$, as these interactions are no longer required.

Under this specification, the estimated effects change noticeably. The point estimates for Phase 1 (IMP1) become substantially larger, while those for Phase 2 (IMP2) are attenuated. It is also important to note the size of this restricted sample: these three metropolitan regions account for the vast majority of the observations in PME, and thus provide substantial identifying power under a more conventional specification. The resulting estimates differ meaningfully in magnitude from those obtained in the baseline specification from Rocha et al. (2018).

Table A11: Rocha et al. (2018) only SP, MG,RJ

	(1)	(2)	(3)
	Being a Formal Entrepreneur		
IMP1	0.0400	0.0413*	0.0284*
	(0.0198)	(0.0194)	(0.0127)
IMP2	0.0491	0.0522	0.0290**
	(0.0264)	(0.0282)	(0.00992)
Basic DD Spec.	Yes	Yes	Yes
Control Econ. Fluc.	No	Yes	Yes
Control Convergence	No	No	Yes
N	22503	22503	22503

Notes: Sample is restricted to the states of Sao Paulo, Minas Gerais and Rio de Janeiro. Specification (1) includes time and entrepreneur fixed effects, as well as individual covariates (age, age squared, and schooling) and interactions with treatment timing. Specification (2) adds controls for local economic conditions and demographics—such as regional wages, hours worked, employment, gender composition, education, and age—along with higher-order terms. Specification (3) further includes convergence controls by interacting the baseline outcome (measured in 2006 at the region–industry level) with polynomial time trends.

Lastly, we re-estimate their specifications without individual fixed effects, treating PME as a repeated cross-section. These results are reported in Table A12. We do so for two reasons. First, PME's rotating panel structure, which implies a short time dimension for each individual, makes two-way fixed effects (TWFE) models with individual fixed effects rely on limited within-person variation over a small number of months. This is precisely the context in which transitions into formal entrepreneurship are relatively infrequent and rotation-induced noise may be non-negligible, potentially reducing statistical power and increasing sensitivity to the panel structure. Second, because treatment varies at the industry $\times$ metropolitan region $\times$ month level and our controls absorb common shocks, a repeated cross-section estimator should recover the same causal contrasts, provided that compositional changes are adequately controlled for.

Comparing the TWFE specification with individual fixed effects to this repeated-cross-section approach therefore provides a natural robustness check: similar estimates indicate that the results are not driven by short-panel idiosyncrasies but instead reflect the intended cell-level difference-in-differences variation. Within this block, Columns (1)–(3) use the full sample, while Columns (4)–(6) apply the same approach to the early-metro subset.

The results reported in Panel B show that the specifications without individual fixed effects yield estimates that differ noticeably from those obtained under the baseline TWFE specification with individual fixed effects. This suggests that the baseline estimates are sensitive to the short rotating panel structure of PME and are not fully replicated when identification instead relies on repeated cross-sectional variation. In other words, the estimated effects of IMP1 change when individual fixed effects are removed, indicating that the original results place substantial weight on within-person variation during the relatively narrow staggered rollout window, rather than on broader cell-level treatment contrasts.

Table A12: Rocha et al. (2018) no Individual Fixed Effects

	(1)	(2)	(3)	(4)	(5)	(6)
	Being a Formal Entrepreneur					
IMP1	0.0257*	0.0265**	0.0199	0.0286*	0.0343**	0.0251*
	(0.00951)	(0.00957)	(0.00986)	(0.0107)	(0.0113)	(0.0117)
IMP2	0.0309*	0.0330*	0.0288*	0.0210	0.0237	0.0152
	(0.0125)	(0.0121)	(0.0119)	(0.0130)	(0.0133)	(0.0129)
Basic DD Spec.	Yes	Yes	Yes	Yes	Yes	Yes
Control Econ. Fluc.	No	Yes	Yes	No	Yes	Yes
Control Convergence	No	No	Yes	No	No	Yes
N	31753	31753	31753	22503	22503	22503

Notes: This table provides estimates of equation 8 without individual fixed effects. Columns (1)–(3) use the full sample, and Columns (4)–(6) apply the same approach to the early-metro subset.

Overall, our findings suggest that individual-level analysis using PME does not appear to offer clear advantages for our research question. Given the coarser industry classification available in PME, which limits the precision of treatment assignment, we therefore rely on PNAD as our primary household survey data source. Nevertheless, the aggregate patterns observed in PME are highly consistent with those documented using PNAD, providing additional reassurance regarding the robustness of our main results.

How the specification issues extend to the RAIS analysis in Rocha et al. (2018)

The considerations discussed above regarding identifying variation in the PME analysis are also informative for understanding the empirical strategy based on RAIS data in Rocha et al. (2018).

Their empirical strategy is very similar to the one described above. Specifically, they estimate equation 10 using a balanced panel at the CNAE × region × time level.

$$
\begin{aligned}
Y_{srt} = {} & \beta_1\big(Post1_{rt} \times Eligible_s\big) + \beta_2\big(Post2_{rt} \times Eligible_s\big) \\
& + \gamma_1 Post1_{rt} + \big(Post1_{rt} \times Eligible_s\big)\gamma_2 \\
& + \eta_s + \lambda_t + \nu_s + \zeta_r + \varepsilon_{srt},
\end{aligned}
\tag{10}
$$

where Y_{srt} denotes the log of the number of formal firms in industry s, region r, and trimester

t. The term $Post1_t$ is a dummy that indicates whether access to IMP1 was available in region r at time t, while $Post2_t$ is the analogous variable for IMP2, which does not vary across regions. The variable $Eligible_s$ denotes the dummy for eligible industries. $Post1_t$ denotes a set of indicator variables for the different start dates of IMP1 across regions, i.e., $t \in \{2\text{nd trim } 2009\}$ and $t \in \{1\text{st trim } 2010\}$, while the variable η_s denotes the interaction between the regional dummies and the time dummies in $Post1_t$. The terms λ_t, ν_s, and ζ_r denote, respectively, time, industry, and region fixed effects.

Here, a similar consideration applies. The inclusion of $\left(Post1_t \times Eligible_s\right)$ leads β_1 to be identifying only using variation from the first months of the program.

Using their data and replication package, we first replicate their results, and then change their specification, similarly to what is described above. Table A13 shows these results. Column (1) shows their original result, column (2) shows the result excluding $Post(Feb2010)_t \times Eligible_s$, column (3) excludes the whole $\left(Post1_t \times Eligible_s\right)$, and in column (4) we restrict our analysis to states for which MEI was available since July 2009 and excluding the vector $\left(Post1_t \times Eligible_s\right)$.

The original interpretation from Rocha et al. (2018) posits that the first phase - allegedly captured by IMP1 in their regression, had no effect on the number of formal firms. Instead, we see that when allowing IMP1 to capture variation after February 2010, these results change substantially, similarly when we look at only those states that had the program since its beginning. These results actually show large effects for the first phase of the program - as we have shown in our main estimates.

Table A13: Rocha et al. (2018) RAIS Analysis with Changes

	(1)	(2)	(3)	(4)
	Log (Number of Formal Firms)			
	Original	No Post Feb2010 x Eligible	No Post x Eligible	Only SP, MG, RJ, DF
IMP1	-0.0106	0.0361***	0.0926***	0.0935***
	(0.00673)	(0.00612)	(0.0114)	(0.0167)
IMP2	0.0426***	0.0478***	0.0475***	0.0717***
	(0.00646)	(0.00674)	(0.00672)	(0.0114)
N	500472	500472	500472	74144

Notes: This table provides estimates of equation 10 using data from Rocha et al. (2018) replication package. Column (1) shows their original result, column (2) shows the result excluding $Post(Feb2010)_t \times Eligible_s$, column (3) excludes the whole $\left(Post1_t \times Eligible_s\right)$, and in column (4) we restrict our analysis to states for which MEI was available since July 2009 and excluding the vector $\left(Post1_t \times Eligible_s\right)$. Sample comprises a balanced panel at the industries x region x quarter level.